

Large-scale animal model study uncovers altered brain pH and lactate levels as a transdiagnostic endophenotype of neuropsychiatric disorders involving cognitive impairment

Hideo Hagihara , Hirotaka Shoji, Satoko Hattori, Giovanni Sala, Yoshihiro Takamiya, Mika Tanaka, Masafumi Ihara, Mihiro Shibutani, Izuho Hatada, Kei Hori, Mikio Hoshino, Akito Nakao, Yasuo Mori, Shigeo Okabe, Masayuki Matsushita, Anja Urbach, Yuta Katayama, Akinobu Matsumoto, Keiichi I. Nakayama, Shota Katori, Takuya Sato, Takuji Iwasato, Haruko Nakamura, Yoshio Goshima, Matthieu Raveau, Tetsuya Tatsukawa, Kazuhiro Yamakawa, Noriko Takahashi, Haruo Kasai, Johji Inazawa, Ikuo Nobuhisa, Tetsushi Kagawa, Tetsuya Taga, Mohamed Darwish, Hirofumi Nishizono, Keizo Takao, Kiran Sapkota, Kazutoshi Nakazawa, Tsuyoshi Takagi, Haruki Fujisawa, Yoshihisa Sugimura, Kyosuke Yamanishi, Lakshmi Rajagopal, Nanette Deneen Hannah, Herbert Y. Meltzer, Tohru Yamamoto, Shuji Wakatsuki, Toshiyuki Araki, Katsuhiko Tabuchi, Tadahiro Numakawa, Hiroshi Kunugi, Freesia L. Huang, Atsuko Hayata-Takano, Hitoshi Hashimoto, Kota Tamada, Toru Takumi, Takaoki Kasahara, Tadafumi Kato, Isabella A. Graef, Gerald R. Crabtree, Nozomi Asaoka, Hikari Hatakama, Shuji Kaneko, Takao Kohno, Mitsuharu Hattori, Yoshio Hoshiba, Ryuhei Miyake, Kisho Obi-Nagata, Akiko Hayashi-Takagi, Léa J. Becker, Ipek Yalcin, Yoko Hagino, Hiroko Kotajima-Murakami, Yuki Moriya, Kazutaka Ikeda, Hyopil Kim, Bong-Kiun Kaang, Hikari Otabi, Yuta Yoshida, Atsushi Toyoda, Noboru H. Komiyama, Seth G. N. Grant, Michiru Ida-Eto, Masaaki Narita, Ken-ichi Matsumoto, Emiko Okuda-Ashitaka, Iori Ohmori, Tadayuki Shimada, Kanato Yamagata, Hiroshi Ageta, Kunihiro Tsuchida, Kaoru Inokuchi, Takayuki Sassa, Akio Kihara, Motoaki Fukasawa, Nobuteru Usuda, Tayo Katano, Teruyuki Tanaka, Yoshihiro Yoshihara, Michihiro Igarashi, Takashi Hayashi, Kaori Ishikawa, Satoshi Yamamoto, Naoya Nishimura, Kazuto Nakada, Shinji Hirotsune, Kiyoshi Egawa, Kazuma Higashisaka, Yasuo Tsutsumi, Shoko Nishihara, Noriyuki Sugo, Takeshi Yagi, Naoto Ueno, Tomomi Yamamoto, Yoshihiro Kubo, Rie Ohashi, Nobuyuki Shiina, Kimiko Shimizu, Sayaka Higo-Yamamoto, Katsutaka Oishi, Hisashi Mori, Tamio Furuse, Masaru Tamura, Hisashi Shirakawa, Daiki X. Sato, Yukiko U. Inoue, Takayoshi Inoue, Yuriko Komine, Tetsuo Yamamori, Kenji Sakimura, Tsuyoshi Miyakawa 

Division of Systems Medical Science, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan • Department of Neurology, National Cerebral and Cardiovascular Center, Suita, Osaka 564-8565, Japan • Laboratory of Genome Science, Biosignal Genome Resource Center, Institute for Molecular and Cellular Regulation, Gunma University, Maebashi, Gunma 371-8512, Japan • Department of Biochemistry and Cellular Biology, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan • Department of Synthetic Chemistry and Biological Chemistry, Graduate School of Engineering, Kyoto University, Kyoto, Kyoto 615-8510, Japan • Department of Cellular Neurobiology, Graduate School of Medicine, The University of Tokyo, Tokyo 113-0033, Japan • Department of Molecular and Cellular Physiology, Graduate School of Medicine, University of the Ryukyus, Nishihara, Okinawa 903-0215, Japan • Hans Berger Department of Neurology, Jena University Hospital, Jena D07747, Germany • Department of Molecular and Cellular Biology, Medical Institute of Bioregulation, Kyushu University, Fukuoka, Fukuoka 812-8582, Japan • Laboratory of Mammalian Neural Circuits, National Institute of Genetics, Mishima, Shizuoka 411-8540, Japan • Department of Molecular Pharmacology and Neurobiology, Yokohama City University Graduate School of Medicine, Yokohama, Kanagawa 236-0004, Japan • Laboratory for Neurogenetics, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan •

Department of Neurodevelopmental Disorder Genetics, Institute of Brain Sciences, Nagoya City University Graduate School of Medical Sciences, Nagoya, Aichi 467-8601, Japan • Laboratory of Structural Physiology, Center for Disease Biology and Integrative Medicine, Faculty of Medicine, The University of Tokyo, Tokyo 113-0033, Japan • Department of Physiology, Kitasato University School of Medicine, Sagamihara, Kanagawa 252-0374, Japan • International Research Center for Neurointelligence (WPI-IRCN), UTIAS, The University of Tokyo, Tokyo 113-0033, Japan • Research Core Center, Tokyo Medical and Dental University, Tokyo 113-8510, Japan • Department of Stem Cell Regulation, Medical Research Institute, Tokyo Medical and Dental University, Tokyo 113-8510, Japan • Department of Behavioral Physiology, Graduate School of Innovative Life Science, University of Toyama, Toyama, Toyama 930-0194, Japan • Department of Biochemistry, Faculty of Pharmacy, Cairo University, Cairo 11562, Egypt • Medical Research Institute, Kanazawa Medical University, Uchinada, Kahoku, Ishikawa 920-0293, Japan • Life Science Research Center, University of Toyama, Toyama, Toyama 930-0194, Japan • Department of Neuroscience, Southern Research, Birmingham, AL 35205, USA • Institute for Developmental Research, Aichi Developmental Disability Center, Kasugai, Aichi 480-0392, Japan • Department of Endocrinology, Diabetes and Metabolism, School of Medicine, Fujita Health University, Toyoake, Aichi 470-1192, Japan • Department of Neuropsychiatry, Hyogo Medical University, School of Medicine, Nishinomiya, Hyogo 663-8501, Japan • Department of Psychiatry and Behavioral Sciences, Northwestern University Feinberg School of Medicine, Chicago, IL 60611, USA • Department of Molecular Neurobiology, Faculty of Medicine, Kagawa University, Kita-gun, Kagawa 761-0793, Japan • Department of Peripheral Nervous System Research, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan • Department of Molecular & Cellular Physiology, Shinshu University School of Medicine, Matsumoto, Nagano 390-8621, Japan • Department of Mental Disorder Research, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan • Department of Psychiatry, Teikyo University School of Medicine, Tokyo 173-8605, Japan • Program of Developmental Neurobiology, National Institute of Child Health and Human Development, National Institute of Health, Bethesda, MD 20892, USA • Laboratory of Molecular Neuropharmacology, Graduate School of Pharmaceutical Sciences, Osaka University, Suita, Osaka 565-0871, Japan • Department of Pharmacology, Graduate School of Dentistry, Osaka University, Suita, Osaka 565-0871, Japan • United Graduate School of Child Development, Osaka University, Kanazawa University, Hamamatsu University School of Medicine, Chiba University and University of Fukui, Suita, Osaka 565-0871, Japan • Division of Bioscience, Institute for Data Biology Science, Osaka University, Suita, Osaka 565-0871, Japan • Transdimensional Life Imaging Division, Institute for Open and Transdisciplinary Research Initiatives, Osaka University, Suita, Osaka 565-0871, Japan • Department of Molecular Pharmaceutical Science, Graduate School of Medicine, Osaka University, Suita, Osaka 565-0871, Japan • RIKEN Brain Science Institute, Wako, Saitama 351-0198, Japan • Department of Physiology and Cell Biology, Kobe University School of Medicine, Kobe, Hyogo 650-0017, Japan • Laboratory for Molecular Dynamics of Mental Disorders, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan • Institute of Biology and Environmental Sciences, Carl von Ossietzky University of Oldenburg, Oldenburg 26129, Germany • Department of Psychiatry and Behavioral Science, Juntendo University Graduate School of Medicine, Tokyo 113-8421, Japan • Department of Pathology, Stanford University School of Medicine, Stanford, CA 94305, USA • Department of Pharmacology, Kyoto Prefectural University of Medicine, Kyoto, Kyoto 602-8566, Japan • Department of Molecular Pharmacology, Graduate School of Pharmaceutical Sciences, Kyoto University, Kyoto, Kyoto 606-8501, Japan • Department of Biomedical Science, Graduate School of Pharmaceutical Sciences, Nagoya City University, Nagoya, Aichi 467-8603, Japan • Laboratory of Medical Neuroscience, Institute for Molecular and Cellular Regulation, Gunma University, Maebashi, Gunma 371-8512, Japan • Laboratory for Multi-scale Biological Psychiatry, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan • Institut des Neurosciences Cellulaires et Intégratives, Centre National de la Recherche Scientifique, Université de Strasbourg, Strasbourg 67000, France • Addictive Substance Project, Tokyo Metropolitan Institute of Medical Science, Tokyo 156-8506, Japan • Department of Biological Sciences, College of Natural Sciences, Seoul National University, Seoul 08826, South Korea • College of Agriculture, Ibaraki University, Ami, Ibaraki 300-0393, Japan • United Graduate School of Agricultural Science, Tokyo University of Agriculture and Technology, Fuchu, Tokyo 183-8538, Japan • Ibaraki University Cooperation between Agriculture and Medical Science (IUCAM), Ami, Ibaraki 300-0393, Japan • Genes to Cognition Program, Centre for Clinical Brain Sciences, University of Edinburgh, Edinburgh EH16 4SB, UK • Simons Initiative for the Developing Brain, Centre for Discovery

Brain Sciences, University of Edinburgh, Edinburgh EH8 9XF, UK • Department of Developmental and Regenerative Medicine, Mie University, Graduate School of Medicine, Tsu, Mie 514-8507, Japan • Department of Biosignaling and Radioisotope Experiment, Interdisciplinary Center for Science Research, Organization for Research and Academic Information, Shimane University, Izumo, Shimane 693-8501, Japan • Department of Biomedical Engineering, Osaka Institute of Technology, Osaka 535-8585, Japan • Department of Physiology, Okayama University Graduate School of Medicine, Dentistry and Pharmaceutical Sciences, Okayama, Okayama 700-8530, Japan • Child brain project, Tokyo Metropolitan Institute of Medical Science, Tokyo 156-8506, Japan • Division for Therapies Against Intractable Diseases, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan • Research Center for Idling Brain Science, University of Toyama, Toyama, Toyama 930-0194, Japan • Department of Biochemistry, Graduate School of Medicine and Pharmaceutical Sciences, University of Toyama, Toyama, Toyama 930-0194, Japan • Core Research for Evolutionary Science and Technology (CREST), Japan Science and Technology Agency (JST), University of Toyama, Toyama, Toyama 930-0194, Japan • Faculty of Pharmaceutical Sciences, Hokkaido University, Sapporo, Hokkaido 060-0812, Japan • Department of Anatomy II, Fujita Health University School of Medicine, Toyoake, Aichi 470-1192, Japan • Department of Medical Chemistry, Kansai Medical University, Hirakata, Osaka 573-1010, Japan • Department of Developmental Medical Sciences, Graduate School of Medicine, The University of Tokyo, Tokyo 113-0033, Japan • Laboratory for Systems Molecular Ethology, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan • Department of Neurochemistry and Molecular Cell Biology, Niigata University Graduate School of Medical and Dental Sciences, Niigata, Niigata 951-8510, Japan • Transdisciplinary Research Program, Niigata University, Niigata, Niigata 951-8510, Japan • Biomedical Research Institute, National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Ibaraki 305-8566, Japan • Faculty of Life and Environmental Sciences, University of Tsukuba, Tsukuba, Ibaraki 305-8572, Japan • Graduate School of Life and Environmental Sciences, University of Tsukuba, Tsukuba, Ibaraki 305-8572, Japan • Integrated Technology Research Laboratories, Pharmaceutical Research Division, Takeda Pharmaceutical Company, Ltd, Fujisawa, Kanagawa 251-8555, Japan • Department of Genetic Disease Research, Osaka City University Graduate School of Medicine, Osaka, Osaka 545-8585, Japan • Department of Pediatrics, Hokkaido University Graduate School of Medicine, Sapporo, Hokkaido 060-8638, Japan • Laboratory of Toxicology and Safety Science, Graduate School of Pharmaceutical Sciences, Osaka University, Suita, Osaka 565-0871, Japan • Glycan & Life Systems Integration Center (GaLSIC), Soka University, Tokyo 192-8577, Japan • Graduate School of Frontier Biosciences, Osaka University, Suita, Osaka 565-0871, Japan • National Institute for Basic Biology, Laboratory of Morphogenesis, Okazaki, Aichi 444-8585, Japan • Division of Biophysics and Neurobiology, National Institute for Physiological Sciences, Okazaki, Aichi 444-8585, Japan • Laboratory of Neuronal Cell Biology, National Institute for Basic Biology, Okazaki, Aichi 444-8585, Japan • Department of Basic Biology, SOKENDAI (Graduate University for Advanced Studies), Okazaki, Aichi 444-8585, Japan • Exploratory Research Center on Life and Living Systems (ExCELLS), Okazaki, Aichi 444-8585, Japan • Department of Biological Sciences, School of Science, The University of Tokyo, Tokyo 113-0033, Japan • Healthy Food Science Research Group, Cellular and Molecular Biotechnology Research Institute, National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Ibaraki 305-8566, Japan • Department of Applied Biological Science, Graduate School of Science and Technology, Tokyo University of Science, Noda, Chiba 278-8510, Japan • Department of Computational Biology and Medical Sciences, Graduate School of Frontier Sciences, The University of Tokyo, Kashiwa, Chiba 277-8562, Japan • School of Integrative and Global Majors (SIGMA), University of Tsukuba, Tsukuba, Ibaraki 305-8577, Japan • Department of Molecular Neuroscience, Graduate School of Medicine and Pharmaceutical Sciences, University of Toyama, Toyama, Toyama 930-0194, Japan • Technology and Development Team for Mouse Phenotype Analysis, Japan Mouse Clinic, RIKEN BioResource Research Center (BRC), Tsukuba, Ibaraki 305-0074, Japan • Graduate School of Life Sciences, Tohoku University, Sendai, Miyagi 980-8578, Japan • Young Researcher Support Group, Research Enhancement Strategy Office, National Institute for Basic Biology, National Institute of Natural Sciences, Okazaki, Aichi 444-8585, Japan • Division of Brain Biology, National Institute for Basic Biology, Okazaki, Aichi 444-8585, Japan • Laboratory for Molecular Analysis of Higher Brain Function, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan • Department of Cellular Neurobiology, Brain Research Institute, Niigata University, Niigata, Niigata 951-8585, Japan • Department of Animal Model Development, Brain Research Institute, Niigata University, Niigata, Niigata 951-8585, Japan

Abstract

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint version 2

December 28, 2023

Reviewed preprint version 1

September 1, 2023 (this version)

Posted to preprint server

June 25, 2023

Sent for peer review

June 25, 2023

Increased levels of lactate, an end-product of glycolysis, have been proposed as a potential surrogate marker for metabolic changes during neuronal excitation. These changes in lactate levels can result in decreased brain pH, which has been implicated in patients with various neuropsychiatric disorders. We previously demonstrated that such alterations are commonly observed in five mouse models of schizophrenia, bipolar disorder, and autism, suggesting a shared endophenotype among these disorders rather than mere artifacts due to medications or agonal state. However, there is still limited research on this phenomenon in animal models, leaving its generality across other disease animal models uncertain. Moreover, the association between changes in brain lactate levels and specific behavioral abnormalities remains unclear. To address these gaps, the International Brain pH Project Consortium investigated brain pH and lactate levels in 109 strains/conditions of 2,294 animals with genetic and other experimental manipulations relevant to neuropsychiatric disorders. Systematic analysis revealed that decreased brain pH and increased lactate levels were common features observed in multiple models of depression, epilepsy, Alzheimer's disease, and some additional schizophrenia models. While certain autism models also exhibited decreased pH and increased lactate levels, others showed the opposite pattern, potentially reflecting subpopulations within the autism spectrum. Furthermore, utilizing large-scale behavioral test battery, a multivariate cross-validated prediction analysis demonstrated that poor working memory performance was predominantly associated with increased brain lactate levels. Importantly, this association was confirmed in an independent cohort of animal models. Collectively, these findings suggest that altered brain pH and lactate levels, which could be attributed to dysregulated excitation/inhibition balance, may serve as transdiagnostic endophenotypes of debilitating neuropsychiatric disorders characterized by cognitive impairment, irrespective of their beneficial or detrimental nature.

eLife assessment

The manuscript offers **useful** descriptive insights into the potential influence of whole-brain lactate and pH levels on the manifestation of behavioral phenotypes seen in diverse animal models of neuropsychiatric disorders. However, reviewers have raised concerns about the potential loss of specificity in capturing regional and cell-type-specific effects when relying solely on whole-brain analysis methods. While the evidence supporting the conclusions is largely **solid**, the robustness of these conclusions could be enhanced by the inclusion of additional data and further analysis.

Introduction

Neuropsychiatric disorders, such as schizophrenia (SZ), bipolar disorder (BD), major depressive disorder (MDD), autism spectrum disorder (ASD), and Alzheimer's disease (AD), are relatively common. More than one-third of the population in most countries is diagnosed with at least one of these disorders at some point in their life ([WHO International Consortium in Psychiatric](#)

Epidemiology, 2000 [\[1\]](#)). Although these diseases are characterized by different clinical diagnostic categories, they share some biological features, such as genetic mutations, molecular changes, and brain activity alterations (Argyelan et al., 2014 [\[2\]](#); Cardno and Owen, 2014 [\[3\]](#); Douaud et al., 2014 [\[4\]](#); Forero et al., 2016 [\[5\]](#); Hall et al., 2015 [\[6\]](#)), suggesting a common underlying biological basis. Accumulating evidence suggests that metabolic changes in the brain are common to several neuropsychiatric disorders. Increased levels of lactate, an end product of the glycolysis pathway, have been observed in the brains of patients with SZ, BD, ASD, MDD, and epilepsy (Dager et al., 2004 [\[7\]](#); Goh et al., 2014 [\[8\]](#); Greene et al., 2003 [\[9\]](#); Halim et al., 2008 [\[10\]](#); Machado-Vieira et al., 2017 [\[11\]](#); Prabakaran et al., 2004 [\[12\]](#); Rossignol and Frye, 2012 [\[13\]](#); Rowland et al., 2016 [\[14\]](#); Soeiro-de-Souza et al., 2016 [\[15\]](#); Sullivan et al., 2019 [\[16\]](#)). Brain lactate levels have been observed to rise during neuronal excitation induced by somatic stimuli (Koush et al., 2019 [\[17\]](#); Mangia et al., 2007 [\[18\]](#)) and epileptic seizures (During et al., 1994 [\[19\]](#); Lazeyras et al., 2000 [\[20\]](#); Najm et al., 1997 [\[21\]](#); Siesjö et al., 1985 [\[22\]](#)). This increase is accompanied by a concurrent elevation in the excitatory neurotransmitter glutamate (Fernandes et al., 2020 [\[23\]](#); Schaller et al., 2014 [\[24\]](#), 2013 [\[25\]](#)). Additionally, lactate has been found to increase the excitability of several populations of neurons (Magistretti and Allaman, 2018 [\[26\]](#)). The presence of increased brain lactate levels aligns with the neuronal hyperexcitation hypothesis proposed for neuropsychiatric disorders, such as SZ (Heckers and Konradi, 2015 [\[27\]](#); Whitfield-Gabrieli et al., 2009 [\[28\]](#)), BD (Chen et al., 2011 [\[29\]](#); Mertens et al., 2015 [\[30\]](#)), and AD (Bi et al., 2020 [\[31\]](#); Busche and Konnerth, 2015 [\[32\]](#)).

Increased lactate levels decrease tissue pH, which may also be associated with deficits in brain energy metabolism (Prabakaran et al., 2004 [\[12\]](#)). Lactate is a relatively strong acid and is almost completely dissociated into H^+ ions and lactate anions at cellular pH (Siesjö, 1985 [\[22\]](#)). Furthermore, H^+ ions are one of the most potent intrinsic neuromodulators in the brain in terms of concentration and thus play an important role in the control of gene expression (Hagihara et al., 2023 [\[33\]](#); Mexal et al., 2006 [\[34\]](#); Mistry and Pavlidis, 2010 [\[35\]](#)) and cellular functions of neurons and glial cells (Chesler, 2003 [\[36\]](#); Kaila and Ransom, 1998 [\[37\]](#)). Recent meta-analyses have confirmed decreased brain pH and increased lactate levels in patients with SZ and BD (Dogan et al., 2018 [\[38\]](#); Pruett and Meador-Woodruff, 2020 [\[39\]](#)). These changes have also been observed in the brains of AD patients (Lehéricy et al., 2007 [\[40\]](#); Liguori et al., 2016 [\[41\]](#), 2015 [\[42\]](#); Lyros et al., 2019 [\[43\]](#); Mullins et al., 2018 [\[44\]](#); Paasila et al., 2019 [\[45\]](#); Youssef et al., 2018 [\[46\]](#)). However, the observed phenomena are potentially confounded by secondary factors inherent to human studies, such as the administration of antipsychotic drugs (Halim et al., 2008 [\[10\]](#)). Agonal experiences associated with these disorders may also complicate the interpretation of postmortem study results (Li et al., 2004 [\[47\]](#); Tomita et al., 2004 [\[48\]](#); Vawter et al., 2006 [\[49\]](#)). Although some human studies have suggested that medication use is not a major factor in regulating brain pH and lactate levels (Dager et al., 2004 [\[7\]](#); Halim et al., 2008 [\[10\]](#); Kato et al., 1998 [\[50\]](#); Machado-Vieira et al., 2017 [\[11\]](#); Soeiro-de-Souza et al., 2016 [\[15\]](#)), it is technically difficult to exclude the effects of other potential confounding factors in such studies, especially those using postmortem brain samples. Animal models are exempt from such confounding factors and may therefore help confirm whether increased brain lactate levels and decreased pH are involved in the pathophysiology of neuropsychiatric and neurodegenerative disorders.

Recently, we reported that increased lactate levels and decreased pH are commonly observed in the postmortem brains of five genetic mouse models of SZ, BD, and ASD (Hagihara et al., 2018 [\[51\]](#)). All mice used in the study were drug-naïve, with equivalent agonal states, postmortem intervals, and ages within each strain. An *in vivo* magnetic resonance spectroscopy (MRS) study showed increased brain lactate levels in another mouse model of SZ (das Neves Duarte et al., 2012 [\[52\]](#)), suggesting that this change is not a postmortem artifact. Thus, these findings in mouse models suggest that increased lactate levels and decreased pH reflect the underlying pathophysiology of the disorders and are not mere artifacts. However, knowledge of brain lactate and especially pH in animal models of neuropsychiatric and neurodegenerative diseases is limited to a small number of models. In particular, few studies have examined brain pH and lactate levels in animal models of MDD, epilepsy, and AD. In addition, few studies other than ours (Hagihara et al., 2018 [\[51\]](#)) have

examined the brain pH and lactate levels in the same samples in relevant animal models. Systematic evaluations using the same platform have not yet been performed. Therefore, given the availability of established relevant animal models, we launched a research project named the “Brain pH Project” (Hagihara et al., 2021b [↗](#)). The aim of this project was to improve our understanding of changes in brain pH, particularly in animal models of neuropsychiatric and neurodegenerative disorders. We have extended our previous small-scale study (Hagihara et al., 2018 [↗](#)) by using a greater variety of animal models of neuropsychiatric disorders and neurodegenerative disorders, including not only SZ, BD, and ASD, but also MDD, epilepsy, AD, and peripheral diseases or conditions comorbid with psychiatric disorders (e.g., diabetes mellitus (DM), colitis, and peripheral nerve injury). These animal models included 109 strains or conditions of mice, rats, and chicks with genetic modifications, drug treatments, and other experimental manipulations such as exposure to physical and psychological stressors. Of these, 65 strains/conditions of animal models constituted an exploratory cohort (Hagihara et al., 2021b [↗](#)) and 44 constituted a confirmatory cohort used to test the hypothesis developed in the initial exploratory studies. We also implemented a statistical learning algorithm that integrated large-scale brain lactate data with comprehensive behavioral measures covering a broad range of behavioral domains (Takao and Miyakawa, 2006 [↗](#)) (e.g., working memory, locomotor activity in a novel environment, sensorimotor gating functions, anxiety-like behavior, and depression-like behavior) to identify behavioral signatures intrinsically related to changes in brain lactate levels. Importantly, by replicating these studies separately in a distinct cohort, we obtained reliable results regarding the potential functional significance of brain lactate changes in animal models of neuropsychiatric disorders.

Results

Altered brain pH and lactate levels in animal models of neuropsychiatric and neurodegenerative disorders

The strains/conditions of animals analyzed in this study and the related diseases/conditions are summarized in Table S1. The raw pH and lactate data and detailed information about the animals (age, sex, and storage duration of tissue samples) are shown in Table S2.

Of the 65 strains/conditions in the exploratory cohort, 26 showed significant changes in pH (6 increased, 20 decreased) and 24 showed significant changes in lactate levels (19 increased, 5 decreased) compared with the corresponding control animals ($P < 0.05$; **Figure 1A** [↗](#), Table S3). No strain/condition of animals showed a concomitant significant increase or decrease in pH and lactate levels. Effect size-based analysis of 65 strains/conditions showed a significant negative correlation between pH and lactate levels at the strain/condition level ($r = -0.86$, $P = 8.45 \times 10^{-20}$; **Figure 1A** [↗](#)). Furthermore, the Z-score-based meta-analysis of 1,239 animals in the exploratory cohort revealed a highly significant negative correlation between pH and lactate levels at the individual animal level ($r = -0.62$, $P = 7.54 \times 10^{-135}$; **Figure 1B** [↗](#)). These results support the idea that decreased brain pH is due to increased lactate levels in pathological conditions associated with neuropsychiatric disorders.

Poor working memory performance predicts higher brain lactate levels

Most of the animal models analyzed have shown a wide range of behavioral abnormalities, such as deficits in learning and memory, increased depression- and anxiety-like behaviors, and impaired sensorimotor gating. Thereafter, with our comprehensive lactate data, we investigated the potential relationship between lactate changes and behavioral phenotypes in animal models. To this end, we examined whether behavioral patterns could predict brain lactate levels by

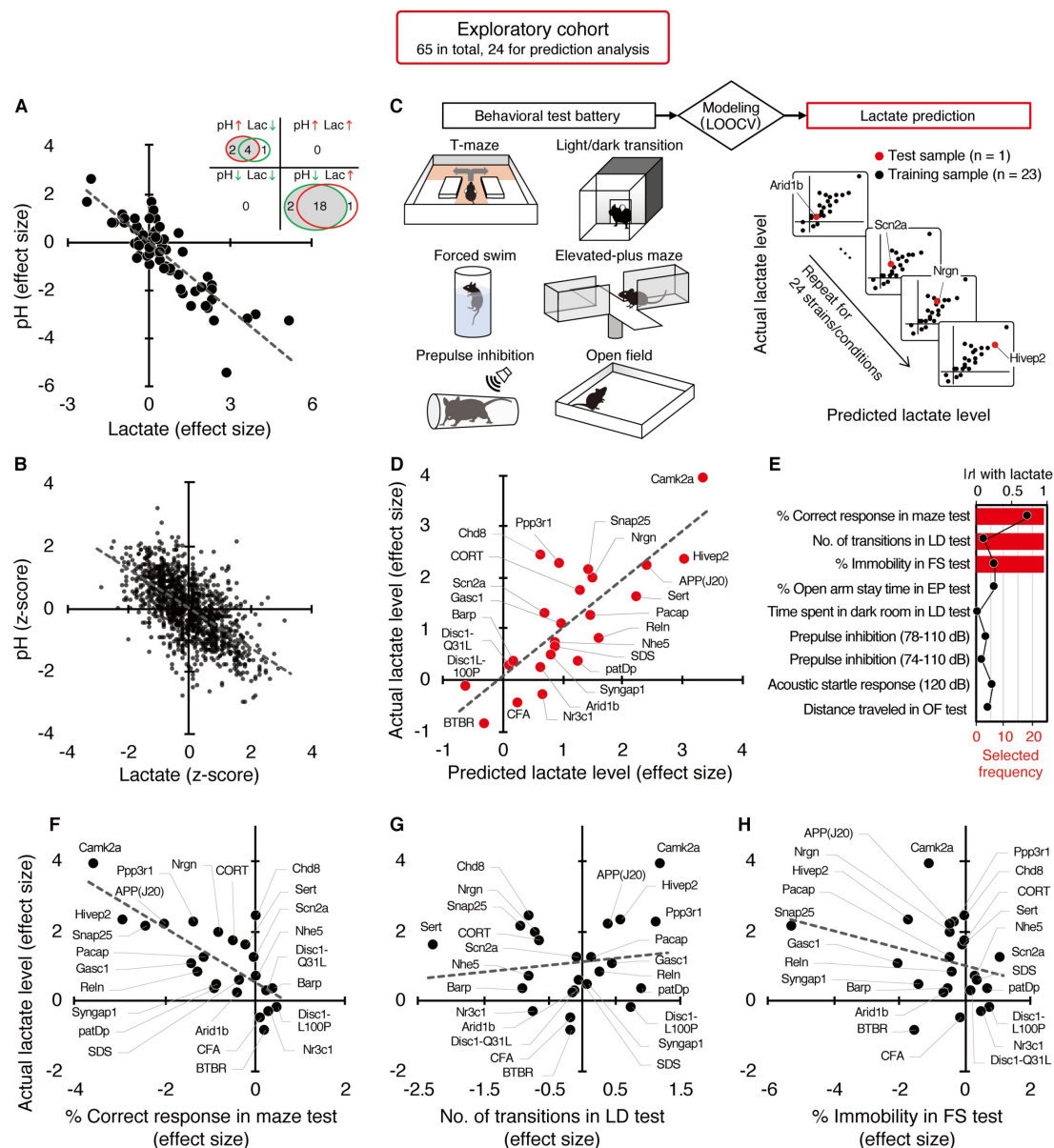


Figure 1.

Increased brain lactate levels correlated with decreased pH are associated with poor working memory.

(A) Venn diagrams show the number of strains/conditions of animal models with significant changes ($P < 0.05$ compared with the corresponding controls) in brain pH and lactate levels in an exploratory cohort. Scatter plot shows the effect size-based correlations between pH and lactate levels of 65 strains/conditions of animals in the cohort. **(B)** Scatter plot showing the z-score-based correlations between pH and lactate levels of 1,239 animals in the cohort. A z-score was calculated for each animal within the strain/condition and used in this study. **(C)** Schematic diagram of the prediction analysis pipeline. Statistical learning models with leave-one-out cross-validation (LOOCV) were built using a series of behavioral data to predict brain lactate levels in 24 strains/conditions of mice in an exploratory cohort. **(D)** The scatter plot shows significant correlations between predicted and actual lactate levels. **(E)** Feature preference for constructing the model to predict brain lactate levels. Bar graphs indicate the selected frequency of behavioral indices in the LOOCV. Line graph indicates absolute correlation coefficient between brain lactate levels and each behavioral measure of the 24 strains/conditions of mice. r , Pearson's correlation coefficient. **(F-H)** Scatter plot showing correlations between actual brain lactate levels and measures of working memory (F), the number of transitions in the light/dark transition test (G), and the percentage of immobility in the forced swim test (H).

applying a statistical learning algorithm to reveal intrinsic associations between brain chemical signatures and behavior. Of the 65 animal models, we collected comprehensive behavioral data from 24 mouse models available from public sources (e.g., published papers and database repositories) and in-house studies (Table S4). We constructed an effect-size-based model to predict brain lactate levels from behavioral data using the leave-one-out cross-validation (LOOCV) method (**Figure 1C**, Table S5). Statistical evaluation of the predictive accuracy of the model revealed a significant correlation between the actual and predicted brain lactate levels ($r = 0.79$, $P = 4.17 \times 10^{-6}$; **Figure 1D**). The calculated root mean square error (RSME) was 0.68. These results indicate that behavioral measures have the potential to predict brain lactate levels in individual models.

Prediction analysis was implemented to evaluate the behavioral measures that were most useful in characterizing the brain lactate levels of the individual strains. The prediction algorithm identified behavioral signatures associated with changes in brain lactate levels by weighting the behavioral measures according to their individual predictive strengths. Thus, we identified behavioral measures associated with changes in brain lactate levels by examining the weighted behavioral measures used for prediction in linear regression. Three out of the nine behavioral measures were selected to build a successful prediction model, and an index of working memory was the top selected measure (**Figure 1E**). According to a simple correlation analysis, working memory measures were significantly negatively correlated with brain lactate levels ($r = -0.76$, $P = 1.93 \times 10^{-5}$; **Figure 1F**). The other two indices used in the successful prediction model did not show significant correlations with brain lactate levels (number of transitions in the light/dark transition test, $r = 0.13$, $P = 0.55$, **Figure 1G**; or percentage of immobility in the forced swim test, $r = -0.28$, $P = 0.19$, **Figure 1H**). Scatter plots of other behavioral indices are shown in Figure S1. Behavioral indices with higher correlation coefficients with actual lactate levels were not necessarily preferentially selected to construct the prediction model (**Figure 1E**). These results suggest that higher brain lactate levels are predominantly linearly related to poorer performance in working memory tests in mouse models of neuropsychiatric disorders. Lactate levels had a V-shape-like relationship with the number of transitions in the light/dark transition test (**Figure 1G**) and the percentage of open-arm stay time in the elevated-plus maze test (Figure S1), which are indices of anxiety-like behavior. This suggests that increased brain lactate levels may also be associated with changes in anxiety-like behaviors, regardless of the direction of the change (increase or decrease).

Validation studies in an independent confirmatory cohort

In a confirmatory cohort consisting of 44 strains/conditions of animal models, 11 strains/conditions showed significant changes in pH (2 increased, 9 decreased) and 11 in lactate levels (10 increased, 1 decreased) compared with the corresponding controls ($P < 0.05$; **Figure 2A**, Table S3). As observed in the exploratory cohort, there were highly significant negative correlations between brain pH and lactate levels at both the strain/condition ($r = -0.78$, $P = 4.07 \times 10^{-10}$; **Figure 2A**) and individual levels ($r = -0.52$, $P = 1.13 \times 10^{-74}$; **Figure 2B**) in this confirmatory cohort.

We then tested the hypothesis developed in the exploratory study that behavioral outcomes predict brain lactate levels. A *priori* power analysis based on an exploratory study ($r = 0.79$, **Figure 1D**) estimated that at least 18 strains/conditions of animals would be required to statistically confirm the results at a level of $\alpha = 0.01$, $|\rho| = 0.79$, $1 - \beta = 0.95$ (Figure S3). Of the 44 strains/conditions of animals in the confirmatory cohort, we collected comprehensive behavioral data from 27 mouse strains from public sources (e.g., published papers and the Mouse Phenotype Database) and unpublished in-house studies (Table S4) that met the criteria for the aforementioned *priori* power analysis. Cross-validation analysis, performed in the same manner as in the exploratory study, showed that behavioral patterns could predict brain lactate levels in the confirmatory cohort ($r = 0.55$, $P = 3.19 \times 10^{-3}$; **Figures 2C** and **2D**). An RMSE value of 0.70 suggests that the prediction accuracy was comparable between the exploratory and confirmatory

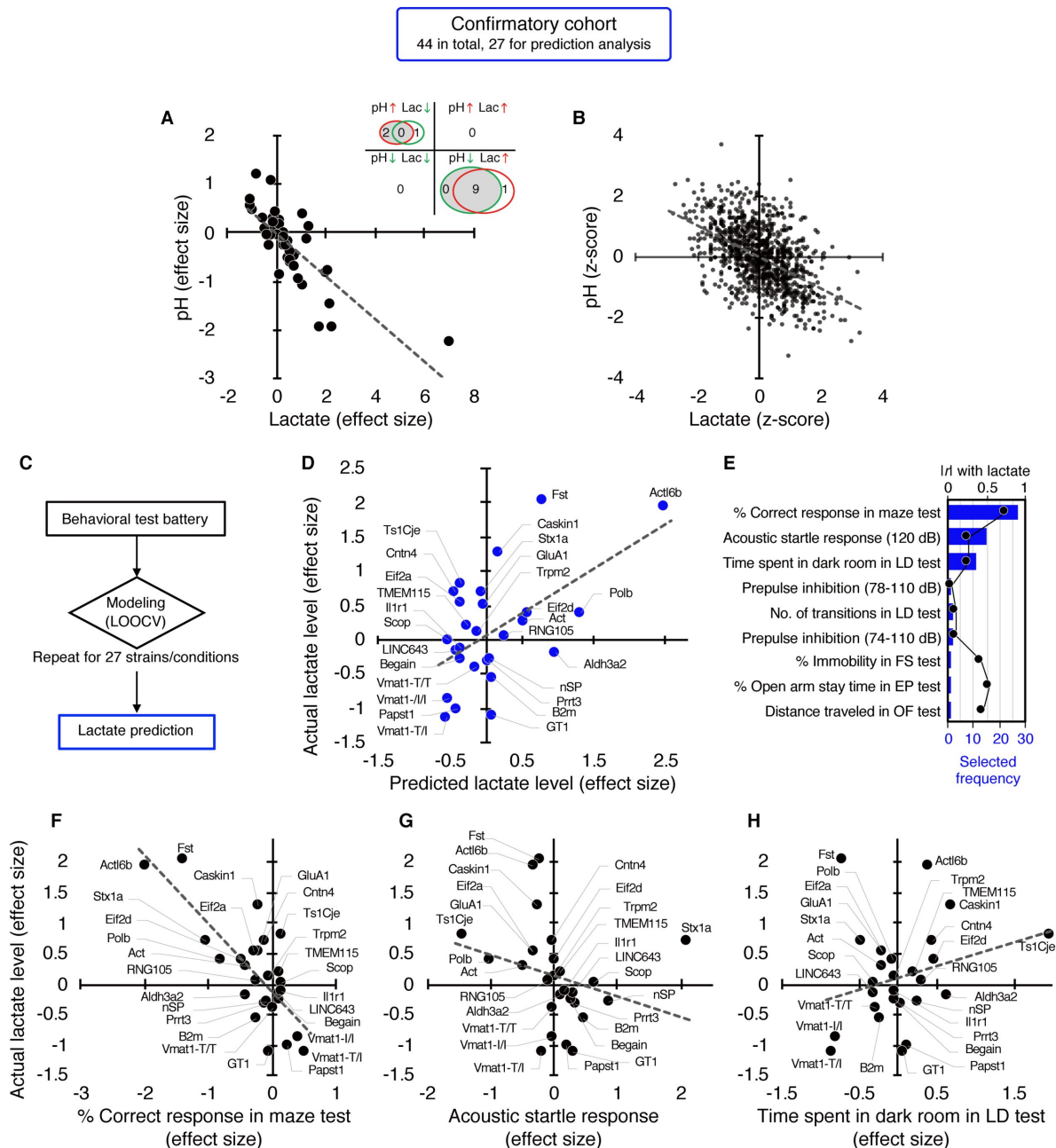


Figure 2.

Studies in an independent confirmatory cohort validate the negative correlation of brain lactate levels with pH and the association of increased lactate with poor working memory.

(A) Venn diagrams show the number of strains/conditions of animal models with significant changes ($P < 0.05$ compared with the corresponding controls) in brain pH and lactate levels in a confirmatory cohort. Scatter plot shows the effect size-based correlations between pH and lactate levels of 44 strains/conditions of animals in the cohort. (B) Scatter plot showing the z-score-based correlations between pH and lactate levels of 1,055 animals in the cohort. (C) Statistical learning models with leave-one-out cross-validation (LOOCV) were built using a series of behavioral data to predict brain lactate levels in 27 strains/conditions of mice in the confirmatory cohort. (D) The scatter plot shows significant correlations between predicted and actual lactate levels. (E) Feature preference for constructing the model to predict brain lactate levels. Bar graphs indicate the selected frequency of behavioral indices in the LOOCV. Line graph indicates absolute correlation coefficient between brain lactate levels and each behavioral index of the 27 strains of mice. r , Pearson's correlation coefficient. (F–H) Scatter plots showing correlations between actual brain lactate levels and working memory measures (F), the acoustic startle response at 120 dB (G), and the time spent in dark room in the light/dark transition test (H).

cohorts (0.68 vs. 0.70, respectively). We found that working memory measures were the most frequently selected behavioral measures for constructing a successful prediction model (**Figure 2E**), which is consistent with the results of the exploratory study (**Figure 1E**). However, other behavioral measures were selected at different frequencies (**Figure 2E**). Simple correlation analyses showed that working memory measures were negatively correlated with brain lactate levels ($r = -0.76$, $P = 6.78 \times 10^{-6}$; **Figure 2F**). No significant correlation with lactate levels was found for the acoustic startle response ($r = -0.26$, $P = 0.21$; **Figure 2G**) or the time spent in the dark room in the light/dark transition test ($r = 0.27$, $P = 0.19$; **Figure 2H**), which were the second and third behavioral measures selected in the prediction model (**Figure 2E**). Again, behavioral indices with higher correlation coefficients were not necessarily preferentially selected to construct the prediction model (**Figure 2E**).

Clustering of 109 strains/conditions of animal models based on changes in brain pH and lactate levels

Combining the exploratory and confirmatory cohorts (109 strains/conditions in total), 37 strains/conditions showed significant changes in pH (8 increased, 29 decreased) and 35 showed significant changes in lactate (29 increased, 6 decreased) compared to the corresponding controls ($P < 0.05$; **Figure S5A**, **Table S3**). Highly significant negative correlations were observed between brain pH and lactate levels at both the strain/condition ($r = -0.80$, $P = 6.99 \times 10^{-26}$; **Figure S5A**) and individual levels ($r = -0.58$, $P = 4.16 \times 10^{-203}$; **Figure S5B**), to a greater extent than observed in each cohort.

In the prediction analysis, behavioral patterns were able to predict brain lactate levels in the combined cohort (51 strains/conditions), as expected (**Figures S5C** and **S5D**, $r = 0.72$, $P = 3.13 \times 10^{-9}$). Furthermore, behavioral patterns predicted brain pH (**Figures S5F** and **S5G**, $r = 0.62$, $P = 9.92 \times 10^{-7}$). In both the lactate and pH prediction models, working memory measures were among the most weighted predictors (**Figures S5E** and **S5H**). Working memory measures were significantly negatively correlated with brain lactate levels and positively correlated with pH (**Figure S6**). Moreover, the number of transitions in the light/dark transition test and the percentage of open arm stay time in the elevated-plus maze test showed a V-shape-like relationship with lactate levels (**Figure S6**).

Hierarchical clustering based on effect size roughly classified all 109 strains/conditions of animals into four groups: low pH/high lactate group, high pH/low lactate group, moderate-high pH/moderate-low lactate group, and a group with minimal to no changes in pH or lactate. These groups consisted of 30, 2, 15, and 62 strains/conditions of animals, respectively (**Figure S7**), where “high” and “low” indicate higher and lower pH and lactate levels in the mutant/experimental animals relative to the corresponding wild-type/control animals, respectively. For example, the low pH/high lactate group included SZ model Nrgn KO mice, SZ/intellectual disability (ID) models Ppp3r1 KO mice and Hivep2 (also known as Shn2) KO mice, AD model APP-J20 Tg mice, ASD model Chd8 KO mice, and social defeat stress-induced depression model mice. Chicks exposed to isolation stress showed decreased brain pH and were included in this group, suggesting that changes in brain pH in response to stress are an interspecies phenomenon. The high to moderate-high pH/low to moderate-low lactate group included mouse models of ASD or developmental delay, such as Shank2 KO, Fmr1 KO, BTBR, Stxbp1 KO, Dyrk1 KO, Aut2 KO, and patDp mice (**Table S1**, **Figure S7**).

Effects of age, sex, and storage duration on brain pH and lactate levels

There was variation among the strains/conditions of the animal models studied with respect to age at sampling, sex, and storage duration of the tissues in the freezer prior to measurements (**Table S2**). We tested the potential effects of these three factors on the brain pH and lactate levels in samples from wild-type and control rodents. Multivariate linear regression analysis using raw pH

values showed that storage duration, but not age or sex, was a significant covariate of brain pH (Figure S8A). None of these three factors covaried with the raw lactate values. Raw pH values were significantly positively correlated with storage duration ($r = 0.11$, $P = 0.00060$; Figure S8D) but not with age ($r = 0.038$, $P = 0.22$; Figure S8B). No significant correlation was observed between the raw lactate values and age ($r = 0.036$, $P = 0.24$) or storage duration ($r = 0.034$, $P = 0.29$) (Figures S8C and S8E). There were no significant differences in pH ($P = 0.42$) or lactate values ($P = 0.22$) between female and male rodents (Figures S8F and S8G).

Discussion

We performed a large-scale analysis of brain pH and lactate levels in 109 animal models of neuropsychiatric disorders, which revealed the diversity of brain energy metabolism among these animal models. Some strains of mice that were considered models of different diseases showed similar patterns of changes in pH and lactate levels. Specifically, the SZ/ID models (Ppp3r1 KO, Nrgn KO mice, and Hivep2 KO mice), BD/ID model (Camk2a KO mice), ASD model (Chd8 KO mice), depression models (mice exposed to social defeat stress, corticosterone-treated mice, and Sert KO mice), AD model (APP-J20 Tg mice), and DM model (Il18 KO and STZ-treated mice) commonly exhibited decreased brain pH and increased lactate levels. BD model Polg1 Tg mice showed no differences in pH or lactate levels. Interestingly, however, other BD model Clock mutant mice and ASD models, such as Shank2 KO (Won et al., 2012), Fmr1 KO, Dyrk1 KO (Raveau et al., 2018), Aut2 KO (Hori et al., 2015), and patDp mice (Nakatani et al., 2009), were classified into a group with opposite changes (a group with decreased lactate levels and increased pH). Animal models with different patterns of changes in brain pH and lactate levels may represent subpopulations of patients or specific disease states (Rossignol and Frye, 2012). While increased brain lactate levels in neuropsychiatric conditions are almost consistent in the literature, decreased lactate levels have also been found in a cohort of patients with SZ (Beasley et al., 2009) and in the euthymic state of BD (Brady et al., 2012). Our results from animal studies may also support the idea that patients classified into specific neuropsychiatric disorders based on symptoms are biologically heterogeneous (Insel and Cuthbert, 2015) from a brain energy metabolism perspective.

Although previous studies have repeatedly reported that brain pH is decreased in SZ and BD (Dogan et al., 2018; Hagihara et al., 2018; Pruett and Meador-Woodruff, 2020), little is known about brain pH in MDD. Our present study demonstrated that decreased brain pH is a common feature in several preclinical animal models of depression (e.g., mice exposed to social defeat stress, corticosterone-treated mice, and Sert KO mice) and comorbid depression (DM mouse model induced by streptozotocin treatment and colitis mouse model induced by dextran sulfate sodium treatment). These findings raise the possibility that decreased brain pH associated with increased lactate levels may be a common endophenotype in MDD, shared with other neuropsychiatric disorders, and needs to be elucidated in future research.

The present animal studies revealed a strong negative correlation between brain pH and lactate levels, which supports our previous findings from small-scale animal studies (Hagihara et al., 2018). A negative correlation between brain pH and lactate levels was found in a human postmortem study (Halim et al., 2008). These results suggest that brain lactate is an important regulator of tissue pH (Prabakaran et al., 2004), although we cannot exclude the possibility that other factors, such as neuronal activity-regulated production of carbon dioxide, another metabolic acid, may also contribute to changes in brain pH (Chesler, 2003; Zauner et al., 1995). Furthermore, the observed pH changes may be due to the dysregulation of neuronal (Li et al., 2022; Pruett et al., 2023) and astroglial (Theparambil et al., 2020) mechanisms of H^+ ion transport and buffering to regulate intracellular and extracellular pH homeostasis, which should be investigated in our model animals.

We observed no significant correlation between brain pH and age in the wild-type/control rodents. In human studies, inconsistent results have been obtained regarding the correlation between brain pH and age; some studies showed no significant correlation (Monoranu et al., 2009 [↗](#); Preece and Cairns, 2003 [↗](#)), whereas others showed a negative correlation (Forester et al., 2010 [↗](#); Harrison et al., 1995 [↗](#)). The effect of sex on brain pH has been inconsistent in human studies (Monoranu et al., 2009 [↗](#); Preece and Cairns, 2003 [↗](#)). Systematic analyses focusing on the effects of age and sex on brain pH in animal models may help explain the inconsistency in human studies.

Our prediction analysis revealed that poorer working memory performance in animal models of neuropsychiatric disorders may be predominantly associated with higher lactate levels, which was reliably confirmed in an independent cohort. Higher lactate levels have been associated with lower cognition in individuals with SZ (Rowland et al., 2016 [↗](#)) and mild cognitive impairment (Weaver et al., 2015 [↗](#)). Based on these observations, abnormal accumulation of lactate would be expected to have a negative impact on brain function, especially memory formation. However, lactate production stimulated by learning tasks has been suggested to be a requisite for memory formation. Lactate production by astrocytic glycogenolysis and its transport to neurons serves as an energy substrate for neuronal activity and is referred to as astrocyte-neuron lactate shuttle (ANLS). Animal studies have shown that pharmacological disruption of learning task-stimulated lactate production and transport via the ANLS immediately before testing impairs memory formation, as assessed by the plus-shaped maze spontaneous alteration task (testing short-term memory) (Newman et al., 2011 [↗](#)) and inhibitory avoidance task (testing long-term memory) (Descalzi et al., 2019 [↗](#); Suzuki et al., 2011 [↗](#)). Collectively, considering that brain lactate levels increase during stimulation in a temporally (and spatially) restricted manner under physiological conditions (Mangia et al., 2007 [↗](#); Schaller et al., 2014 [↗](#)), pathologically sustained elevation of brain lactate levels may have negative effects on brain functions, including memory processing, although causality is unknown. Another possibility is that the reduced consumption of lactate for energy production due to mitochondrial dysfunction in neurons may underlie impaired learning and memory functions in disease conditions. Mitochondrial dysfunction is thought to lead to lactate accumulation due to the insufficient capacity of mitochondrial metabolism to metabolize the lactate produced (Dogan et al., 2018 [↗](#); Regenold et al., 2009 [↗](#); Stork and Renshaw, 2005 [↗](#)). Mitochondrial dysfunction has been consistently implicated in several neuropsychiatric disorders, including SZ, BD, MDD, ASD, and AD (Holper et al., 2019 [↗](#); Manji et al., 2012 [↗](#); Pei and Wallace, 2018 [↗](#)), among which working memory deficits are a common symptom (Millan et al., 2012 [↗](#)). In addition, increased lactate levels reflect neuronal activation (Hagihara et al., 2018 [↗](#)). Thus, activation in brain regions other than the frontal cortex, a brain region critical for working memory (Andrés, 2003 [↗](#)), interferes with working memory performance, as it has been proposed that the activity of the core brain region may be affected by noise from the rest of the brain during cognitive tasks in patients with SZ (Foucher et al., 2005 [↗](#)). Moreover, increased lactate may have a positive or beneficial effect on memory function to compensate for its impairment, as lactate administration with an associated increase in brain lactate levels attenuates cognitive deficits in human patients (Bisri et al., 2016 [↗](#)) and rodent models (Rice et al., 2002 [↗](#)) of traumatic brain injury. In addition, lactate administration exerts antidepressant effects in a mouse model of depression (Carrard et al., 2016 [↗](#)). Moreover, increased lactate levels may also be involved in behavioral changes other than memory deficits such as anxiety. The results of our previous study showed that increased brain lactate levels were associated with altered anxiety-like behaviors in a social defeat stress model of depression (Hagihara et al., 2021a [↗](#)). Further studies are needed to address these hypotheses by chronically inducing deficits in mitochondrial function to manipulate endogenous lactate levels in a brain region-specific manner and to analyze their effects on working memory.

There exists a close relationship between neuronal activity and energy metabolism in the brain. In vitro studies have indicated that the uptake of glutamate into astrocytes stimulates glycolysis and lactate production following neuronal excitation (Pellerin and Magistretti, 1994 [↗](#)). However, an in

vivo investigation on cerebellar Purkinje cells has demonstrated that lactate is produced in neurons in an activity-dependent manner, suggesting that astrocytes may not be the sole supplier of lactate to neurons (Caesar et al., 2008 [DOI](#)). Shifts in the neuronal excitation and inhibition (E/I) balance toward excitation of specific neural circuits have been implicated in the pathogenesis and pathophysiology of various neuropsychiatric disorders, including SZ, BD, ASD, AD, and epilepsy (Brealy et al., 2015 [DOI](#); Busche and Konnerth, 2016 [DOI](#); Marín, 2012 [DOI](#); Nelson and Valakh, 2015 [DOI](#); Yizhar et al., 2011 [DOI](#)). An imbalance favoring excitation could lead to increased energy expenditure and potentially heightened glycolysis. Such alterations in energy metabolism may be associated with increased lactate production. Indeed, in our previous studies using Hivep2 KO mice, characterized by increased brain lactate levels and decreased pH, we observed elevated glutamate levels and upregulated expression of many glycolytic genes in the hippocampus (Hagihara et al., 2018 [DOI](#); Takao et al., 2013 [DOI](#)). Furthermore, Actl6b (also known as Baf53b) KO mice (Wenderski et al., 2020 [DOI](#)) and APP-J20 Tg mice (Bomben et al., 2014 [DOI](#); Brown et al., 2018 [DOI](#); Palop et al., 2007 [DOI](#)) exhibited neuronal hyperexcitation, as evidenced by increased expression of activity-regulated genes and epileptiform discharges recorded by electroencephalography. Dravet syndrome model mice with a clinically relevant SCN1A mutation (Scn1a-A1783V knock-in mice) (Ricobaraza et al., 2019 [DOI](#)) and mutant Snap25 (S187A) knock-in mice (Kataoka et al., 2011 [DOI](#)) developed convulsive seizures. These findings suggest that the observed increase in lactate production and subsequent decrease in pH in whole-brain samples may be attributed to the hyperactivity of specific neural circuits in a subset of the examined animal models.

Because we used whole brain samples to measure pH and lactate levels, we could not determine whether the observed changes in pH and/or lactate levels occurred ubiquitously throughout the brain or selectively in specific brain region(s) in each strain/condition of the models. Indeed, brain region-specific increases in lactate levels were observed in human patients with ASD in an MRS study (Goh et al., 2014 [DOI](#)). Furthermore, while increased lactate levels were observed in whole-brain measurements in mice with chronic social defeat stress (Figure S7) (Hagihara et al., 2021a [DOI](#)), decreased lactate levels were found in the dorsomedial prefrontal cortex (Yao et al., 2023 [DOI](#)). The brain region-specific changes may occur even in animal models in which undetectable changes were observed in the present study. This could be due to the masking of such changes in the analysis when using whole-brain samples. Further studies are needed to address this issue by measuring microdissected brain samples and performing *in vivo* analyses using pH- or lactate-sensitive biosensor electrodes (Marunaka et al., 2014 [DOI](#); Newman et al., 2011 [DOI](#)) and MRS (Davidovic et al., 2011 [DOI](#)).

In conclusion, the present study demonstrated that altered brain pH and lactate levels are commonly observed in animal models of SZ, BD, ID, ASD, AD, and other neuropsychiatric disorders. These findings provide further evidence supporting the hypothesis that altered brain pH and lactate levels are not mere artifacts, such as those resulting from medication confounding, but are rather involved in the underlying pathophysiology of some patients with neuropsychiatric disorders. Altered brain energy metabolism or neural hyper- or hypoactivity leading to abnormal lactate levels and pH may serve as a potential therapeutic targets for neuropsychiatric disorders (Pruett and Meador-Woodruff, 2020 [DOI](#)). In addition, detection of brain lactate changes by methods such as MRS may help diagnose and subcategorize these biologically heterogeneous disorders, as has been shown for mitochondrial diseases (Lin et al., 2003 [DOI](#)). Future studies are needed to identify effective treatment strategies specific to sets of animal models that could recapitulate the diversity of brain energy metabolism in human disease conditions. This could contribute to the development of treatments for biologically defined subgroups of patients or disease states of debilitating diseases beyond clinically defined boundaries.

Materials and methods

Experimental animals and ethical statement

The animals used in this study are listed in Table S1. Animal experiments were approved by the Institutional Animal Care and Use Committee of Fujita Health University and the relevant committee at each participating institution, based on the Law for the Humane Treatment and Management of Animals and the Standards Relating to the Care and Management of Laboratory Animals and Relief of Pain. Every effort was made to minimize the number of animals used.

Sample collection

Whole brain samples were collected by one of the following methods:

1. Call for collaborative research worldwide, e.g. by posting on the website of the relevant scientific society (<https://www.ibngs.org/news>) and of our institute (http://www.fujita-hu.ac.jp/~cgbb/en/collaborative_research/index.html), and by discussion on a preprint server, *bioRxiv* (<https://www.biorxiv.org/content/10.1101/2021.02.02.428362v2>).
2. Ask specifically the researchers who have established animal models.
3. Purchase or transfer mouse strains of interest from the repository (e.g., The Jackson Laboratory [<https://www.jax.org/>], RIKEN BioResource Research Center [https://web.brc.riken.jp/en/?doing_wp_cron=1612919495.41057896614074707031 25]).
4. Rederivation of mouse strains of interest from frozen embryo stocks.

Sampling and handling of brain samples

We have established a standardized protocol for the sampling and handling of brain samples to minimize potential confounding effects due to technical differences between laboratories and to conduct blinded studies (http://www.fujita-hu.ac.jp/~cgbb/en/collaborative_research/index.html).

Animals and samples

- Animals: Mice, rats, and other laboratory animals. For genetically engineered animals, mutants and their wild-type littermates should be used.
- Number of animals: ≥ 6 per group (identical genetic background, littermate), preferably.
- Sex of animals: All males, all females, or balanced among groups if mixed.
- Samples: Fresh-frozen whole brain.

Blinded study

The pH and lactate measurements were blinded: Upon sampling, the investigators were supposed to randomize the animals regarding genotype and collect the brain samples in serially numbered tubes. The investigators were asked to provide the genotype information and corresponding serial numbers after the measurements for subsequent statistical analyses.

Brain sampling procedures

1. Sacrifice the mouse/rat by cervical dislocation followed by decapitation and remove the entire brain from the skull. Do not immerse the brain in buffer solutions or water.
2. Cut the brain along the longitudinal fissure of the cerebrum.
3. Collect the left and right hemispheres in a tube that can be tightly capped like a cryotube and seal the caps with Parafilm (to minimize the effect of carbon dioxide from dry ice on the tissue pH during transport).

4. Quick freeze in liquid nitrogen and store at -80°C until shipped.
5. Transport the frozen brain on dry ice.

Measurements of pH and lactate

pH and lactate were measured as previously described (Hagihara et al., 2018 [DOI](#)). Briefly, snap-frozen tissues were homogenized in ice-cold distilled H_2O (5 ml per 500 mg of tissue). The pH of the homogenates was measured using a pH meter (LAQUA F-72, HORIBA, Ltd., Kyoto, Japan) equipped with a Micro ToupH electrode (9618S-10D, HORIBA, Ltd.) after three-point calibration at pH 4.0, pH 7.0, and pH 9.0. The concentration of lactate in the homogenates was determined using a multi-assay analyzer (GM7 MicroStat, Analox Instruments, London, UK) after calibration with 8.0 M lactate standard solution (GMRD-103, Analox Instruments). A 20- μl aliquot of the centrifuged supernatant (14,000 rpm, 10 min) was used for the measurement.

Effect size (d) was calculated for each strain/condition and each measure (i.e., pH, lactate value, and behavioral index) as followed:

$$d = (M_{\text{mutants}} - M_{\text{controls}}) / S_{\text{pooled}}$$

$$S_{\text{pooled}} = [(S^2_{\text{mutant}} + S^2_{\text{control}}) / 2]^{1/2}$$

The heat map was depicted using the R (version 3.5.2) gplots package.

Z-score transformation, a traditional method of data normalization for direct comparison between different samples and conditions, was applied to each pH or lactate value using individual animal data within each of strain according to the following formula:

$$\text{Z-score} = (\text{value}_P - \text{mean value}_{P1...Pn}) / \text{standard deviation}_{P1...Pn},$$

where P is any pH or lactate and P1...Pn represent the aggregate measure of all pH or lactate values.

Prediction analysis

We collected the comprehensive behavioral data as much as of animal models whose brain pH and lactate levels were examined in this study. We obtained the following behavioral data from 24 animal models in an exploratory cohort from published papers, the Mouse Phenotype Database (<http://www.mouse-phenotype.org/> [DOI](#)) or in-house studies (Tables S3 and S4): number of transitions in the light-dark transition test, percentage of immobility in the forced swim test, time spent in open arm in the elevated-plus maze test, prepulse inhibition at 78-110 dB and 74-110 dB, startle response at 120 dB, distance traveled in the open field test, and correct percentage in the T-maze, Y-maze, or eight-arm radial maze test. Literature searches were performed in PubMed and Google Scholar using relevant keywords: name of strain or experimental condition, species (mice or rats), and name of behavioral tests. Among the top hits of the search, data presented as actual values of mean and SD or SEM were used with priority. For some behavioral measures, possible mean and SD values were estimated from the graph presented in the paper. In the matrix of strains/conditions and behavioral measures, those with any missing values were excluded, resulting in nine behavioral measures from 24 strains/conditions of mouse models. The effect size was calculated for each strain/condition and measure and used in the prediction analysis.

Leave-one-out cross-validation was employed to determine whether behavioral measures could predict brain lactate levels for individual mouse strains. From the analyzed behavioral dataset of 24 mouse strains, one sample was selected and excluded to serve as the test data of the cross-validation. Then, a multivariate linear regression model was trained on the remaining 23 samples

using a stepwise variable selection procedure with EZR software (version 1.38; Saitama Medical Center, Jichi Medical University, Saitama, Japan) (Kanda, 2013 [DOI](#)), and the test sample was predicted. This was repeated 24 times, with all samples selected once as the test data. Behavioral measures selected at least once in the prediction model were considered as predictive behavioral measures. Prediction performance was analyzed by evaluating the correlation between predicted and actual values for the 24 mouse strains.

For comparability, we performed prediction analyses in a confirmatory cohort using the nine behavioral indices mentioned above, resulting in the inclusion of 27 strains/conditions of animals (Tables S3 and S4). In the prediction analyses, the same settings as used in an exploratory cohort were applied to the confirmatory and combined cohorts.

To compare prediction accuracy across cohorts, the root mean squared error (RMSE) was calculated using the following formula:

$$\text{RMSE} = \left[\frac{1}{n} \sum_{i=1}^n (f_i - y_i)^2 \right]^{1/2}$$

where n is the total number of samples, f_i is the predicted value, and y_i is the actual value.

Statistical analysis

pH and lactate data were analyzed by unpaired t-test or one-way analysis of variance (ANOVA) or two-way ANOVA followed by *post hoc* Tukey's multiple comparison test using GraphPad Prism 8 (version 8.4.2; GraphPad Software, San Diego, CA). Correlation analysis was performed using Pearson's correlation coefficient method.

Acknowledgements

This work was supported by MEXT KAKENHI (Grant No. JP16H06462 [to T. Miyakawa]), MEXT Promotion of Distinctive Joint Research Center Program (Grant No. JPMXP0618217663 [to T. Miyakawa]), JSPS KAKENHI (Grant Nos. JP20H00522 and JP16H06276 (AdAMS) [to T. Miyakawa]; Grant Nos. JP18K07378 and JP21K19314 [to H. Hagihara]), and AMED Strategic Research Program for Brain Sciences (Grant No. JP18dm0107101 [to T. Miyakawa]).

The International Brain pH Consortium

Division of Systems Medical Science, Center for Medical Science, Fujita Health University, Toyoake, Japan: Hideo Hagihara, Hirotaka Shoji, Giovanni Sala, Satoko Hattori, Daiki X. Sato, Yoshihiro Takamiya, Mika Tanaka, and Tsuyoshi Miyakawa

Department of Neurology, National Cerebral and Cardiovascular Center, Suita, Japan: Masafumi Ihara

Laboratory of Genome Science, Biosignal Genome Resource Center, Institute for Molecular and Cellular Regulation, Gunma University, Maebashi, Japan: Mihiro Shibutani and Izuho Hatada

Department of Biochemistry and Cellular Biology, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Japan: Kei Hori and Mikio Hoshino

Department of Synthetic Chemistry and Biological Chemistry, Graduate School of Engineering, Kyoto University, Kyoto, Japan: Akito Nakao and Yasuo Mori

Department of Cellular Neurobiology, Graduate School of Medicine, The University of Tokyo, Tokyo, Japan: Shigeo Okabe

Department of Molecular and Cellular Physiology, Graduate School of Medicine, University of the Ryukyus, Nishihara, Japan: Masayuki Matsushita

Hans Berger Department of Neurology, Jena University Hospital, Jena, Germany: Anja Urbach

Department of Molecular and Cellular Biology, Medical Institute of Bioregulation, Kyushu University, Fukuoka, Japan: Yuta Katayama, Akinobu Matsumoto, and Keiichi I. Nakayama

Laboratory of Mammalian Neural Circuits, National Institute of Genetics, Mishima, Japan: Shota Katori, Takuya Sato, and Takuji Iwasato

Department of Molecular Pharmacology and Neurobiology, Yokohama City University Graduate School of Medicine, Yokohama, Japan: Haruko Nakamura and Yoshio Goshima

Laboratory for Neurogenetics, RIKEN Center for Brain Science, Wako, Japan: Matthieu Raveau, Tetsuya Tatsukawa, and Kazuhiro Yamakawa

Department of Neurodevelopmental Disorder Genetics, Institute of Brain Sciences, Nagoya City University Graduate School of Medical Sciences, Nagoya, Japan: Kazuhiro Yamakawa

Laboratory of Structural Physiology, Center for Disease Biology and Integrative Medicine, Faculty of Medicine, The University of Tokyo, Tokyo, Japan: Noriko Takahashi and Haruo Kasai

Department of Physiology, Kitasato University School of Medicine, Sagamihara, Japan: Noriko Takahashi

International Research Center for Neurointelligence (WPI-IRCN), UTIAS, The University of Tokyo, Tokyo, Japan: Haruo Kasai

Research Core Center, Tokyo Medical and Dental University, Tokyo, Japan: Johji Inazawa

Department of Stem Cell Regulation, Medical Research Institute, Tokyo Medical and Dental University, Tokyo, Japan: Ikuo Nobuhisa, Tetsushi Kagawa, Tetsuya Taga

Department of Behavioral Physiology, Graduate School of Innovative Life Science, University of Toyama, Toyama, Japan: Mohamed Darwish and Keizo Takao

Department of Biochemistry, Faculty of Pharmacy, Cairo University, Cairo, Egypt: Mohamed Darwish

Medical Research Institute, Kanazawa Medical University, Uchinada, Japan: Hirofumi Nishizono

Life Science Research Center, University of Toyama, Toyama, Japan: Keizo Takao

Department of Neuroscience, Southern Research, Birmingham, AL, USA: Kiran Sapkota and Kazutoshi Nakazawa

Institute for Developmental Research, Aichi Developmental Disability Center, Kasugai, Japan: Tsuyoshi Takagi

Department of Endocrinology, Diabetes and Metabolism, School of Medicine, Fujita Health University, Toyoake, Japan: Haruki Fujisawa and Yoshihisa Sugimura

Department of Neuropsychiatry, Hyogo Medical University, School of Medicine, Nishinomiya, Japan: Kyosuke Yamanishi

Department of Psychiatry and Behavioral Sciences, Northwestern University Feinberg School of Medicine, Chicago, IL, USA: Lakshmi Rajagopal, Nanette Deneen Hannah, and Herbert Y. Meltzer

Department of Molecular Neurobiology, Faculty of Medicine, Kagawa University, Kita-gun, Japan: Tohru Yamamoto

Department of Peripheral Nervous System Research, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Japan: Shuji Wakatsuki, and Toshiyuki Araki

Department of Molecular & Cellular Physiology, Shinshu University School of Medicine, Matsumoto, Japan: Katsuhiko Tabuchi

Department of Mental Disorder Research, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Japan: Tadahiro Numakawa and Hiroshi Kunugi

Department of Psychiatry, Teikyo University School of Medicine, Tokyo, Japan: Hiroshi Kunugi

Program of Developmental Neurobiology, National Institute of Child Health and Human Development, National Institute of Health, Bethesda, MD, USA: Freesia L. Huang

Laboratory of Molecular Neuropharmacology, Graduate School of Pharmaceutical Sciences, Osaka University, Suita, Japan: Atsuko Hayata-Takano and Hitoshi Hashimoto

Department of Pharmacology, Graduate School of Dentistry, Osaka University, Suita, Japan: Atsuko Hayata-Takano

United Graduate School of Child Development, Osaka University, Kanazawa University, Hamamatsu University School of Medicine, Chiba University and University of Fukui, Suita, Japan: Atsuko Hayata-Takano and Hitoshi Hashimoto

Division of Bioscience, Institute for Data Biology Science, Osaka University, Suita, Japan: Hitoshi Hashimoto

Transdimensional Life Imaging Division, Institute for Open and Transdisciplinary Research Initiatives, Osaka University, Suita, Japan: Hitoshi Hashimoto

Department of Molecular Pharmaceutical Science, Graduate School of Medicine, Osaka University, Suita, Japan: Hitoshi Hashimoto

RIKEN Brain Science Institute, Wako, Japan: Kota Tamada and Toru Takumi

Department of Physiology and Cell Biology, Kobe University School of Medicine, Kobe, Japan: Kota Tamada and Toru Takumi

Laboratory for Molecular Dynamics of Mental Disorders, RIKEN Center for Brain Science, Wako, Japan: Takaoki Kasahara and Tadaaki Kato

Institute of Biology and Environmental Sciences, Carl von Ossietzky University of Oldenburg, Oldenburg, Germany: Takaoki Kasahara

Department of Psychiatry and Behavioral Science, Juntendo University Graduate School of Medicine, Tokyo, Japan: Tadafumi Kato

Department of Pathology, Stanford University School of Medicine, Stanford, CA, USA: Isabella A. Graef and Gerald R. Crabtree

Department of Pharmacology, Kyoto Prefectural University of Medicine, Kyoto, Japan: Nozomi Asaoka

Department of Molecular Pharmacology, Graduate School of Pharmaceutical Sciences, Kyoto University, Kyoto, Japan: Hikari Hatakama and Shuji Kaneko

Department of Biomedical Science, Graduate School of Pharmaceutical Sciences, Nagoya City University, Nagoya, Japan: Takao Kohno and Mitsuharu Hattori

Laboratory of Medical Neuroscience, Institute for Molecular and Cellular Regulation, Gunma University, Maebashi, Japan: Yoshio Hoshiba and Akiko Hayashi-Takagi

Laboratory for Multi-scale Biological Psychiatry, RIKEN Center for Brain Science, Wako, Japan: Ryuhei Miyake, Kisho Obi-Nagata, and Akiko Hayashi-Takagi

Institut des Neurosciences Cellulaires et Intégratives, Centre National de la Recherche Scientifique, Université de Strasbourg, Strasbourg, France: Léa J. Becker and Ipek Yalcin

Addictive Substance Project, Tokyo Metropolitan Institute of Medical Science, Tokyo, Japan: Yoko Hagino, Hiroko Kotajima-Murakami, Yuki Moriya, and Kazutaka Ikeda

Department of Biological Sciences, College of Natural Sciences, Seoul National University, Seoul, South Korea: Hyopil Kim and Bong-Kiun Kaang

College of Agriculture, Ibaraki University, Ami, Japan: Hikari Otabi, Yuta Yoshida, and Atsushi Toyoda

United Graduate School of Agricultural Science, Tokyo University of Agriculture and Technology, Fuchu, Japan: Hikari Otabi and Atsushi Toyoda

Ibaraki University Cooperation between Agriculture and Medical Science (IUCAM), Ami, Japan: Atsushi Toyoda

Genes to Cognition Program, Centre for Clinical Brain Sciences, University of Edinburgh, Edinburgh, UK: Noboru H. Komiyama and Seth G. N. Grant

Simons Initiative for the Developing Brain, Centre for Discovery Brain Sciences, University of Edinburgh, Edinburgh, UK: Noboru H. Komiyama and Seth G. N. Grant

Department of Developmental and Regenerative Medicine, Mie University, Graduate School of Medicine, Tsu, Japan: Michiru Ida-Eto and Masaaki Narita

Department of Biosignaling and Radioisotope Experiment, Interdisciplinary Center for Science Research, Organization for Research and Academic Information, Shimane University, Izumo, Japan: Ken-ichi Matsumoto

Department of Biomedical Engineering, Osaka Institute of Technology, Osaka, Japan: Emiko Okuda-Ashitaka

Department of Physiology, Okayama University Graduate School of Medicine, Dentistry and Pharmaceutical Sciences, Okayama, Japan: Iori Ohmori

Child brain project, Tokyo Metropolitan Institute of Medical Science, Tokyo, Japan: Tadayuki Shimada and Kanato Yamagata

Division for Therapies Against Intractable Diseases, Center for Medical Science, Fujita Health University, Toyoake, Japan: Hiroshi Ageta and Kunihiro Tsuchida

Research Center for Idling Brain Science, University of Toyama, Toyama, Japan: Kaoru Inokuchi

Department of Biochemistry, Graduate School of Medicine and Pharmaceutical Sciences, University of Toyama, Toyama, Japan: Kaoru Inokuchi

Core Research for Evolutionary Science and Technology (CREST), Japan Science and Technology Agency (JST), University of Toyama, Toyama, Japan: Kaoru Inokuchi

Faculty of Pharmaceutical Sciences, Hokkaido University, Sapporo, Japan: Takayuki Sassa and Akio Kihara

Department of Anatomy II, Fujita Health University School of Medicine, Toyoake, Japan: Motoaki Fukasawa and Nobuteru Usuda

Department of Medical Chemistry, Kansai Medical University, Hirakata, Japan: Tayo Katano

Department of Developmental Medical Sciences, Graduate School of Medicine, The University of Tokyo, Tokyo, Japan: Teruyuki Tanaka

Laboratory for Systems Molecular Ethology, RIKEN Center for Brain Science, Wako, Japan: Yoshihiro Yoshihara

Department of Neurochemistry and Molecular Cell Biology, Niigata University Graduate School of Medical and Dental Sciences, Niigata, Japan: Michihiro Igarashi

Transdisciplinary Research Program, Niigata University, Niigata, Japan: Michihiro Igarashi

Biomedical Research Institute, National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Japan: Takashi Hayashi

Faculty of Life and Environmental Sciences, University of Tsukuba, Tsukuba, Japan: Kaori Ishikawa and Kazuto Nakada

Graduate School of Life and Environmental Sciences, University of Tsukuba, Tsukuba, Japan: Kaori Ishikawa and Kazuto Nakada

Integrated Technology Research Laboratories, Pharmaceutical Research Division, Takeda Pharmaceutical Company, Ltd, Fujisawa, Japan: Satoshi Yamamoto and Naoya Nishimura

Department of Genetic Disease Research, Osaka City University Graduate School of Medicine, Osaka, Japan: Shinji Hirotsune

Department of Pediatrics, Hokkaido University Graduate School of Medicine, Sapporo, Japan:
Kiyoshi Egawa

Laboratory of Toxicology and Safety Science, Graduate School of Pharmaceutical Sciences, Osaka
University, Suita, Japan: Kazuma Higashisaka and Yasuo Tsutsumi

Glycan & Life Systems Integration Center (GaLSIC), Soka University, Tokyo, Japan: Shoko Nishihara

Graduate School of Frontier Biosciences, Osaka University, Suita, Japan: Noriyuki Sugo and Takeshi
Yagi

National Institute for Basic Biology, Laboratory of Morphogenesis, Okazaki, Japan: Naoto Ueno

Division of Biophysics and Neurobiology, National Institute for Physiological Sciences, Okazaki,
Japan: Tomomi Yamamoto and Yoshihiro Kubo

Laboratory of Neuronal Cell Biology, National Institute for Basic Biology, Okazaki, Japan: Rie
Ohashi and Nobuyuki Shiina

Department of Basic Biology, SOKENDAI (Graduate University for Advanced Studies), Okazaki,
Japan: Rie Ohashi and Nobuyuki Shiina

Exploratory Research Center on Life and Living Systems (ExCELLS), Okazaki, Japan: Rie Ohashi
and Nobuyuki Shiina

Department of Biological Sciences, School of Science, The University of Tokyo, Tokyo, Japan:
Kimiko Shimizu

Healthy Food Science Research Group, Cellular and Molecular Biotechnology Research Institute,
National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Japan: Sayaka
Higo-Yamamoto and Katsutaka Oishi

Department of Applied Biological Science, Graduate School of Science and Technology, Tokyo
University of Science, Noda, Japan: Katsutaka Oishi

Department of Computational Biology and Medical Sciences, Graduate School of Frontier Sciences,
The University of Tokyo, Kashiwa, Japan: Katsutaka Oishi

School of Integrative and Global Majors (SIGMA), University of Tsukuba, Tsukuba, Japan:
Katsutaka Oishi

Department of Molecular Neuroscience, Graduate School of Medicine and Pharmaceutical
Sciences, University of Toyama, Toyama, Japan: Hisashi Mori

Technology and Development Team for Mouse Phenotype Analysis, Japan Mouse Clinic, RIKEN
BioResource Research Center (BRC), Tsukuba, Japan: Tamio Furuse and Masaru Tamura

Department of Molecular Pharmacology, Graduate School of Pharmaceutical Sciences, Kyoto
University, Kyoto, Japan: Hisashi Shirakawa

Graduate School of Life Sciences, Tohoku University, Sendai, Japan: Daiki X. Sato

Department of Biochemistry and Cellular Biology, National Institute of Neuroscience, National
Center of Neurology and Psychiatry, Kodaira, Japan: Yukiko U. Inoue and Takayoshi Inoue

Young Researcher Support Group, Research Enhancement Strategy Office, National Institute for Basic Biology, National Institute of Natural Sciences, Okazaki, Japan: Yuriko Komine

Division of Brain Biology, National Institute for Basic Biology, Okazaki, Japan: Tetsuo Yamamori

Laboratory for Molecular Analysis of Higher Brain Function, RIKEN Center for Brain Science, Wako, Japan: Tetsuo Yamamori

Department of Cellular Neurobiology, Brain Research Institute, Niigata University, Niigata, Japan: Kenji Sakimura

Department of Animal Model Development, Brain Research Institute, Niigata University, Niigata, Japan: Kenji Sakimura

Author contributions

H. Hagihara and T.M. designed the study. H. Hagihara, H. Shoji, S. Hattori, Y. Takamiya, M. Tanaka, M. Ihara, M.S., I.H., K. Hori, M. Hoshino, A.N., Y. Mori, S.O., M.M., A.U., Y. Katayama, A.M., K.I.N., S. Katori, T. Sato, T. Iwasato, H. Nakamura, Y.G., M.R., T. Tatsukawa, K. Yamakawa, N.T., H. Kasai, J.I., I.N., T. Kagawa, T. Taga, M.D., H. Nishizono, K. Takao, K. Sapkota, K. Nakazawa, T. Takagi, H.F., Y.S., K. Yamanishi, L.R., N.D.H., H.Y.M., Tohru Yamamoto, S.W., T.A., K. Tabuchi, T.N., H. Kunugi, F.L.H., A. Hayata-Takano, H. Hashimoto, K. Tamada, T. Takumi, T. Kasahara, T. Kato, I.A.G., G.R.C., N.A., H. Hatakama, S. Kaneko, T. Kohno, M. Hattori, Y.H., R.M., K.O-N., A. Hayashi-Takagi, L.J.B., I.Y., Y.H., H.K-M., Y. Moriya, K. Ikeda, H.K., B-K.K., H.O., Y.Y., A.T., N.H.K., S.G.N.G., M.I-E., M.N., K.M., E.O-A., I.O., T. Shimada, K. Yamagata, H.A., K. Tsuchida, K. Inokuchi, T. Sassa, A.K., M.F., N. Usuda, T. Katano, T. Tanaka, Y.Y., M. Igarashi, T.H., K. Ishikawa, S.Y., N.N., K. Nakada, S. Hirotsune, K.E., K. Higashisaka, Y. Tsutsumi, S.N., N. Sugo, T. Yagi, N. Ueno, Tomomi Yamamoto, Y. Kubo, N. Shiina, R.O., K. Shimizu, S.H-Y., K.O., H.M., T.F., M. Tamura, H. Shirakawa, D.X.S., Y.U.I., T. Inoue, Y. Komine, T. Yamamori, K. Sakimura, and T. Miyakawa prepared animals and tissues. H. Hagihara and G.S. performed biochemical and statistical analyses. H. Hagihara and T. Miyakawa wrote the manuscript. All authors approved the final version of the manuscript.

Disclosures

The authors report no biomedical financial interests or potential conflicts of interest.

Data availability

The data analyzed in this study are available in Supplementary tables.

References

- Andrés P (2003) **Frontal cortex as the central executive of working memory: time to revise our view** *Cortex* **39**:871–895 [https://doi.org/10.1016/S0010-9452\(08\)70868-2](https://doi.org/10.1016/S0010-9452(08)70868-2)
- Argyelan M, Ikuta T, DeRosse P, Braga RJ, Burdick KE, John M, Kingsley PB, Malhotra AK, Szeszko PR (2014) **Resting-state fMRI connectivity impairment in schizophrenia and bipolar disorder** *Schizophr Bull* **40**:100–110 <https://doi.org/10.1093/schbul/sbt092>
- Beasley CL, Dwork AJ, Rosoklija G, Mann JJ, Mancevski B, Jakovski Z, Davceva N, Tait AR, Straus SK, Honer WG (2009) **Metabolic abnormalities in fronto-striatal-thalamic white matter tracts in schizophrenia** *Schizophr Res* **109**:159–166 <https://doi.org/10.1016/j.schres.2009.01.017>
- Bi D, Wen L, Wu Z, Shen Y (2020) **GABAergic dysfunction in excitatory and inhibitory (E/I) imbalance drives the pathogenesis of Alzheimer’s disease** *Alzheimers Dement* **16**:1312–1329 <https://doi.org/10.1002/alz.12088>
- Bisri T, Utomo BA, Fuadi I (2016) **Exogenous lactate infusion improved neurocognitive function of patients with mild traumatic brain injury** *Asian J Neurosurg* **11**:151–159 <https://doi.org/10.4103/1793-5482.145375>
- Bomben V, Holth J, Reed J, Cramer P, Landreth G, Noebels J (2014) **Bexarotene reduces network excitability in models of Alzheimer’s disease and epilepsy** *Neurobiol Aging* **35**:2091–2095 <https://doi.org/10.1016/j.neurobiolaging.2014.03.029>
- Brady RO, Cooper A, Jensen JE, Tandon N, Cohen B, Renshaw P, Keshavan M, Öngür D (2012) **A longitudinal pilot proton MRS investigation of the manic and euthymic states of bipolar disorder** *Transl Psychiatry* **2**:e160–e160 <https://doi.org/10.1038/tp.2012.84>
- Brealy JA, Shaw A, Richardson H, Singh KD, Muthukumaraswamy SD, Keedwell PA (2015) **Increased visual gamma power in schizoaffective bipolar disorder** *Psychol Med* **45**:783–794 <https://doi.org/10.1017/S0033291714001846>
- Brown R *et al.* (2018) **Circadian and brain state modulation of network hyperexcitability in Alzheimer’s disease** *eNeuro* **5**:426–17 <https://doi.org/10.1523/ENEURO.0426-17.2018>
- Busche MA, Konnerth A (2016) **Impairments of neural circuit function in Alzheimer’s disease** *Philos Trans R Soc B Biol Sci* **371** <https://doi.org/10.1098/rstb.2015.0429>
- Busche MA, Konnerth A (2015) **Neuronal hyperactivity – A key defect in Alzheimer’s disease?** *BioEssays* **37**:624–632 <https://doi.org/10.1002/bies.201500004>
- Caesar K, Hashemi P, Douhou A, Bonvento G, Boutelle MG, Walls AB, Lauritzen M (2008) **Glutamate receptor-dependent increments in lactate, glucose and oxygen metabolism evoked in rat cerebellum in vivo** *J Physiol* **586**:1337–1349 <https://doi.org/10.1113/jphysiol.2007.144154>
- Cardno AG, Owen MJ (2014) **Genetic relationships between schizophrenia, bipolar disorder, and schizoaffective disorder** *Schizophr Bull* **40**:504–515 <https://doi.org/10.1093/schbul/sbu016>

- Carrard A, Elsayed M, Margineanu M, Boury-Jamot B, Fragnière L, Meylan EM, Petit J-M, Fiumelli H, Magistretti PJ, Martin J-L (2016) **Peripheral administration of lactate produces antidepressant-like effects** *Mol Psychiatry* **23**:392–399 <https://doi.org/10.1038/mp.2016.179>
- Chen C-H, Suckling J, Lennox BR, Ooi C, Bullmore ET (2011) **A quantitative meta-analysis of fMRI studies in bipolar disorder** *Bipolar Disord* **13**:1–15 <https://doi.org/10.1111/j.1399-5618.2011.00893.x>
- Chesler M (2003) **Regulation and modulation of pH in the brain** *Physiol Rev* **83**:1183–1221 <https://doi.org/10.1152/physrev.00010.2003>
- Dager SR, Friedman SD, Parow A, Demopulos C, Stoll AL, Lyoo IK, Dunner DL, Renshaw PF (2004) **Brain metabolic alterations in medication-free patients with bipolar disorder** *Arch Gen Psychiatry* **61**:450–458 <https://doi.org/10.1001/archpsyc.61.5.450>
- das Neves Duarte JM, Kulak A, Gholam-Razaei MM, Cuenod M, Gruetter R, Do KQ (2012) **N-Acetylcysteine normalizes neurochemical changes in the glutathione-deficient schizophrenia mouse model during development** *Biol Psychiatry, Glutamatergic Treatments for Psychiatric Disorders* **71**:1006–1014 <https://doi.org/10.1016/j.biopsych.2011.07.035>
- Davidovic L, Navratil V, Bonaccorso CM, Catania MV, Bardoni B, Dumas M-E (2011) **A metabolomic and systems biology perspective on the brain of the Fragile X syndrome mouse model** *Genome Res* **21**:2190–2202 <https://doi.org/10.1101/gr.116764.110>
- Descalzi G, Gao V, Steinman MQ, Suzuki A, Alberini CM (2019) **Lactate from astrocytes fuels learning-induced mRNA translation in excitatory and inhibitory neurons** *Commun Biol* **2**:1–11 <https://doi.org/10.1038/s42003-019-0495-2>
- Dogan AE, Yuksel C, Du F, Chouinard V-A, Öngür D. (2018) **Brain lactate and pH in schizophrenia and bipolar disorder: a systematic review of findings from magnetic resonance studies** *Neuropsychopharmacology* **43**:1681–1690 <https://doi.org/10.1038/s41386-018-0041-9>
- Douaud G *et al.* (2014) **A common brain network links development, aging, and vulnerability to disease** *Proc Natl Acad Sci* **111**:17648–17653 <https://doi.org/10.1073/pnas.1410378111>
- During MJ, Fried I, Leone P, Katz A, Spencer DD (1994) **Direct measurement of extracellular lactate in the human hippocampus during spontaneous seizures** *J Neurochem* **62**:2356–2361 <https://doi.org/10.1046/j.1471-4159.1994.62062356.x>
- Fernandes CC, Lanz B, Chen C, Morris PG (2020) **Measurement of brain lactate during visual stimulation using a long TE semi-LASER sequence at 7 T** *NMR Biomed* **33** <https://doi.org/10.1002/nbm.4223>
- Forero DA, Herteleer L, De Zutter S, Norrback K-F, Nilsson L-G, Adolfsson R, Callaerts P, Del-Favero J. (2016) **A network of synaptic genes associated with schizophrenia and bipolar disorder** *Schizophr Res* **172**:68–74 <https://doi.org/10.1016/j.schres.2016.02.012>
- Forester BP *et al.* (2010) **Age-related changes in brain energetics and phospholipid metabolism** *NMR Biomed* **23**:242–250 <https://doi.org/10.1002/nbm.1444>

- Foucher JR, Vidailhet P, Chanraud S, Gounot D, Grucker D, Pins D, Damsa C, Danion J-M (2005) **Functional integration in schizophrenia: too little or too much? Preliminary results on fMRI data** *NeuroImage* **26**:374–388 <https://doi.org/10.1016/j.neuroimage.2005.01.042>
- Goh S, Dong Z, Zhang Y, DiMauro S, Peterson BS (2014) **Brain imaging evidence that mitochondrial dysfunction is a neurobiological subtype of Autism Spectrum Disorder** *JAMA Psychiatry* **71**:665–671 <https://doi.org/10.1001/jamapsychiatry.2014.179>
- Greene AE, Todorova MT, Seyfried TN (2003) **Perspectives on the metabolic management of epilepsy through dietary reduction of glucose and elevation of ketone bodies** *J Neurochem* **86**:529–537 <https://doi.org/10.1046/j.1471-4159.2003.01862.x>
- Hagihara H *et al.* (2018) **Decreased brain pH as a shared endophenotype of psychiatric disorders** *Neuropsychopharmacology* **43**:459–468 <https://doi.org/10.1038/npp.2017.167>
- Hagihara H, Murano T, Miyakawa T (2023) **The gene expression patterns as surrogate indices of pH in the brain** *Front Psychiatry* **14** <https://doi.org/10.3389/fpsy.2023.1151480>
- Hagihara H, Shoji H, Otabi H, Toyoda A, Katoh K, Namihira M, Miyakawa T (2021) **Protein lactylation induced by neural excitation** *Cell Rep* **37** <https://doi.org/10.1016/j.celrep.2021.109820>
- Hagihara H *et al.* (2021) **Systematic analysis of brain lactate and pH levels in 65 animal models related to neuropsychiatric conditions** *bioRxiv* **2021**
- Halim ND, Lipska BK, Hyde TM, Deep-Soboslay A, Saylor EM, Herman M, Thakar J, Verma A, Kleinman JE (2008) **Increased lactate levels and reduced pH in postmortem brains of schizophrenics: medication confounds** *J Neurosci Methods* **169**:208–213 <https://doi.org/10.1016/j.jneumeth.2007.11.017>
- Hall J, Trent S, Thomas KL, O'Donovan MC, Owen MJ (2015) **Genetic risk for schizophrenia: convergence on synaptic pathways involved in plasticity** *Biol Psychiatry, The New Psychiatric Genetics: Toward Next Generation Diagnosis and Treatment* **77**:52–58 <https://doi.org/10.1016/j.biopsych.2014.07.011>
- Harrison PJ, Heath PR, Eastwood SL, Burnet PWJ, McDonald B, Pearson RCA (1995) **The relative importance of premortem acidosis and postmortem interval for human brain gene expression studies: selective mRNA vulnerability and comparison with their encoded proteins** *Neurosci Lett* **200**:151–154 [https://doi.org/10.1016/0304-3940\(95\)12102-A](https://doi.org/10.1016/0304-3940(95)12102-A)
- Heckers S, Konradi C (2015) **GABAergic mechanisms of hippocampal hyperactivity in schizophrenia** *Schizophr Res, The GABA System in Schizophrenia: Cells, Molecules and Microcircuitry* **167**:4–11 <https://doi.org/10.1016/j.schres.2014.09.041>
- Holper L, Ben-Shachar D, Mann JJ (2019) **Multivariate meta-analyses of mitochondrial complex I and IV in major depressive disorder, bipolar disorder, schizophrenia, Alzheimer disease, and Parkinson disease** *Neuropsychopharmacology* **44**:837–849 <https://doi.org/10.1038/s41386-018-0090-0>
- Hori K, Nagai T, Shan W, Sakamoto A, Abe M, Yamazaki M, Sakimura K, Yamada K, Hoshino M (2015) **Heterozygous disruption of autism susceptibility candidate 2 causes impaired emotional control and cognitive memory** *PLOS ONE* **10** <https://doi.org/10.1371/journal.pone.0145979>

- Insel TR, Cuthbert BN (2015) **Brain disorders? Precisely** *Science* **348**:499–500 <https://doi.org/10.1126/science.aab2358>
- Kaila K, Ransom BR (1998) **pH and Brain Function**
- Kanda Y. (2013) **Investigation of the freely available easy-to-use software “EZR” for medical statistics** *Bone Marrow Transplant* **48**:452–458 <https://doi.org/10.1038/bmt.2012.244>
- Kataoka M *et al.* (2011) **A single amino acid mutation in SNAP-25 induces anxiety-related behavior in mouse** *PLOS ONE* **6** <https://doi.org/10.1371/journal.pone.0025158>
- Kato T, Murashita J, Kamiya A, Shioiri T, Kato N, Inubushi T (1998) **Decreased brain intracellular pH measured by 31P-MRS in bipolar disorder: a confirmation in drug-free patients and correlation with white matter hyperintensity** *Eur Arch Psychiatry Clin Neurosci* **248**:301–306 <https://doi.org/10.1007/s004060050054>
- Koush Y, de Graaf RA, Jiang L, Rothman DL, Hyder F. (2019) **Functional MRS with J-edited lactate in human motor cortex at 4LJT** *NeuroImage* **184**:101–108 <https://doi.org/10.1016/j.neuroimage.2018.09.008>
- Lazeyras F, Blanke O, Zimine I, Delavelle J, Perrig SH, Seeck M (2000) **MRI, 1H-MRS, and functional MRI during and after prolonged nonconvulsive seizure activity** *Neurology* **55**:1677–1682 <https://doi.org/10.1212/WNL.55.11.1677>
- Lehéricy S, Marjanska M, Mesrob L, Sarazin M, Kinkingnehun S (2007) **Magnetic resonance imaging of Alzheimer’s disease** *Eur Radiol* **17**:347–362 <https://doi.org/10.1007/s00330-006-0341-z>
- Li JZ *et al.* (2004) **Systematic changes in gene expression in postmortem human brains associated with tissue pH and terminal medical conditions** *Hum Mol Genet* **13**:609–616 <https://doi.org/10.1093/hmg/ddh065>
- Li Y, Fan C, Wang C, Wang L, Yi Y, Mao X, Chen X, Lan T, Wang W, Yu SY (2022) **Stress-induced reduction of Na⁺/H⁺ exchanger isoform 1 promotes maladaptation of neuroplasticity and exacerbates depressive behaviors** *Sci Adv* **8** <https://doi.org/10.1126/sciadv.add7063>
- Liguori C, Chiaravalloti A, Sancesario G, Stefani A, Sancesario GM, Mercuri NB, Schillaci O, Pierantozzi M (2016) **Cerebrospinal fluid lactate levels and brain [18F]FDG PET hypometabolism within the default mode network in Alzheimer’s disease** *Eur J Nucl Med Mol Imaging* **43**:2040–2049 <https://doi.org/10.1007/s00259-016-3417-2>
- Liguori C, Stefani A, Sancesario G, Sancesario GM, Marciani MG, Pierantozzi M (2015) **CSF lactate levels, τ proteins, cognitive decline: a dynamic relationship in Alzheimer’s disease** *J Neurol Neurosurg Psychiatry* **86**:655–659 <https://doi.org/10.1136/jnnp-2014-308577>
- Lin DDM, Crawford TO, Barker PB (2003) **Proton MR spectroscopy in the diagnostic evaluation of suspected mitochondrial disease** *Am J Neuroradiol* **24**:33–41
- Lyros E *et al.* (2019) **Normal brain aging and Alzheimer’s disease are associated with lower cerebral pH: an in vivo histidine 1H-MR spectroscopy study** *Neurobiol Aging* **87**:60–69 <https://doi.org/10.1016/j.neurobiolaging.2019.11.012>

- Machado-Vieira R *et al.* (2017) **Increased brain lactate during depressive episodes and reversal effects by lithium monotherapy in drug-naïve bipolar disorder: A 3T 1H-MRS study** *J Clin Psychopharmacol* **37**:40–45 <https://doi.org/10.1097/JCP.0000000000000616>
- Magistretti PJ, Allaman I (2018) **Lactate in the brain: from metabolic end-product to signalling molecule** *Nat Rev Neurosci* **19**:235–249 <https://doi.org/10.1038/nrn.2018.19>
- Mangia S, Tkáč I, Gruetter R, Van de Moortele P-F, Maraviglia B, Uğurbil K. (2007) **Sustained neuronal activation raises oxidative metabolism to a new steady-state level: evidence from 1H NMR spectroscopy in the human visual cortex** *J Cereb Blood Flow Metab* **27**:1055–1063 <https://doi.org/10.1038/sj.jcbfm.9600401>
- Manji H, Kato T, Di Prospero NA, Ness S, Beal MF, Krams M, Chen G. (2012) **Impaired mitochondrial function in psychiatric disorders** *Nat Rev Neurosci* **13**:293–307 <https://doi.org/10.1038/nrn3229>
- Marín O (2012) **Interneuron dysfunction in psychiatric disorders** *Nat Rev Neurosci* **13**:107–120 <https://doi.org/10.1038/nrn3155>
- Marunaka Y, Yoshimoto K, Aoi W, Hosogi S, Ikegaya H (2014) **Low pH of interstitial fluid around hippocampus of the brain in diabetic OLETF rats** *Mol Cell Ther* **2** <https://doi.org/10.1186/2052-8426-2-6>
- Mertens J *et al.* (2015) **Differential responses to lithium in hyperexcitable neurons from patients with bipolar disorder** *Nature* **527**:95–99 <https://doi.org/10.1038/nature15526>
- Mexal S, Berger R, Adams CE, Ross RG, Freedman R, Leonard S (2006) **Brain pH has a significant impact on human postmortem hippocampal gene expression profiles** *Brain Res* **1106**:1–11 <https://doi.org/10.1016/j.brainres.2006.05.043>
- Millan MJ *et al.* (2012) **Cognitive dysfunction in psychiatric disorders: characteristics, causes and the quest for improved therapy** *Nat Rev Drug Discov* **11**:141–168 <https://doi.org/10.1038/nrd3628>
- Mistry M, Pavlidis P (2010) **A cross-laboratory comparison of expression profiling data from normal human postmortem brain** *Neuroscience* **167**:384–395 <https://doi.org/10.1016/j.neuroscience.2010.01.016>
- Monoranu CM *et al.* (2009) **pH measurement as quality control on human post mortem brain tissue: a study of the BrainNet Europe consortium** *Neuropathol Appl Neurobiol* **35**:329–337 <https://doi.org/10.1111/j.1365-2990.2008.01003a.x>
- Mullins R, Reiter D, Kapogiannis D (2018) **Magnetic resonance spectroscopy reveals abnormalities of glucose metabolism in the Alzheimer’s brain** *Ann Clin Transl Neurol* **5**:262–272 <https://doi.org/10.1002/acn3.530>
- Najm IM, Wang Y, Hong SC, Luders HO, Ng TC, Comair YG (1997) **Temporal changes in proton MRS metabolites after kainic acid-induced seizures in rat brain** *Epilepsia* **38**:87–94 <https://doi.org/10.1111/j.1528-1157.1997.tb01082.x>
- Nakatani J *et al.* (2009) **Abnormal behavior in a chromosome-engineered mouse model for human 15q11-13 duplication seen in autism** *Cell* **137**:1235–1246 <https://doi.org/10.1016/j.cell.2009.04.024>

Nelson SB, Valakh V (2015) **Excitatory/inhibitory balance and circuit homeostasis in autism spectrum disorders** *Neuron* **87**:684–698 <https://doi.org/10.1016/j.neuron.2015.07.033>

Newman LA, Korol DL, Gold PE (2011) **Lactate produced by glycogenolysis in astrocytes regulates memory processing** *PLOS ONE* **6** <https://doi.org/10.1371/journal.pone.0028427>

Paasila PJ, Davies DS, Kril JJ, Goldsbury C, Sutherland GT (2019) **The relationship between the morphological subtypes of microglia and Alzheimer’s disease neuropathology** *Brain Pathol* **29**:726–740 <https://doi.org/10.1111/bpa.12717>

Palop JJ *et al.* (2007) **Aberrant excitatory neuronal activity and compensatory remodeling of inhibitory hippocampal circuits in mouse models of Alzheimer’s disease** *Neuron* **55**:697–711 <https://doi.org/10.1016/j.neuron.2007.07.025>

Pei L, Wallace DC (2018) **Mitochondrial etiology of neuropsychiatric disorders** *Biol Psychiatry, Mitochondrial Dynamics and Bioenergetics in Psychiatry* **83**:722–730 <https://doi.org/10.1016/j.biopsych.2017.11.018>

Pellerin L, Magistretti PJ (1994) **Glutamate uptake into astrocytes stimulates aerobic glycolysis: a mechanism coupling neuronal activity to glucose utilization** *Proc Natl Acad Sci* **91**:10625–10629

Prabakaran S *et al.* (2004) **Mitochondrial dysfunction in schizophrenia: evidence for compromised brain metabolism and oxidative stress** *Mol Psychiatry* **9**:684–697 <https://doi.org/10.1038/sj.mp.4001511>

Preece P, Cairns NJ (2003) **Quantifying mRNA in postmortem human brain: influence of gender, age at death, postmortem interval, brain pH, agonal state and inter-lobe mRNA variance** *Mol Brain Res* **118**:60–71 [https://doi.org/10.1016/S0169-328X\(03\)00337-1](https://doi.org/10.1016/S0169-328X(03)00337-1)

Pruett BS, Meador-Woodruff JH (2020) **Evidence for altered energy metabolism, increased lactate, and decreased pH in schizophrenia brain: A focused review and meta-analysis of human postmortem and magnetic resonance spectroscopy studies** *Schizophr Res* **223**:29–42 <https://doi.org/10.1016/j.schres.2020.09.003>

Pruett BS, Pinner AL, Kim P, Meador-Woodruff JH (2023) **Altered distribution and localization of organellar Na⁺/H⁺ exchangers in postmortem schizophrenia dorsolateral prefrontal cortex** *Transl Psychiatry* **13**:1–10 <https://doi.org/10.1038/s41398-023-02336-2>

Raveau M, Shimohata A, Amano K, Miyamoto H, Yamakawa K (2018) **DYRK1A-haploinsufficiency in mice causes autistic-like features and febrile seizures** *Neurobiol Dis* **110**:180–191 <https://doi.org/10.1016/j.nbd.2017.12.003>

Regenold WT, Phatak P, Marano CM, Sassan A, Conley RR, Kling MA (2009) **Elevated cerebrospinal fluid lactate concentrations in patients with bipolar disorder and schizophrenia: implications for the mitochondrial dysfunction hypothesis** *Biol Psychiatry, Impaired Neuroplasticity in Schizophrenia* **65**:489–494 <https://doi.org/10.1016/j.biopsych.2008.11.010>

Rice AC, Zsoldos R, Chen T, Wilson MS, Alessandri B, Hamm RJ, Ross Bullock M (2002) **Lactate administration attenuates cognitive deficits following traumatic brain injury** *Brain Res* **928**:156–159 [https://doi.org/10.1016/S0006-8993\(01\)03299-1](https://doi.org/10.1016/S0006-8993(01)03299-1)

- Ricobaraza A *et al.* (2019) **Epilepsy and neuropsychiatric comorbidities in mice carrying a recurrent Dravet syndrome SCN1A missense mutation** *Sci Rep* **9** <https://doi.org/10.1038/s41598-019-50627-w>
- Rossignol DA, Frye RE (2012) **Mitochondrial dysfunction in autism spectrum disorders: a systematic review and meta-analysis** *Mol Psychiatry* **17**:290–314 <https://doi.org/10.1038/mp.2010.136>
- Rowland LM, Pradhan S, Korenic S, Wijtenburg SA, Hong LE, Edden RA, Barker PB (2016) **Elevated brain lactate in schizophrenia: a 7 T magnetic resonance spectroscopy study** *Transl Psychiatry* **6**:e967–e967 <https://doi.org/10.1038/tp.2016.239>
- Schaller B, Mekle R, Xin L, Kunz N, Gruetter R (2013) **Net increase of lactate and glutamate concentration in activated human visual cortex detected with magnetic resonance spectroscopy at 7 tesla** *J Neurosci Res* **91**:1076–1083 <https://doi.org/10.1002/jnr.23194>
- Schaller B, Xin L, O'Brien K, Magill AW, Gruetter R (2014) **Are glutamate and lactate increases ubiquitous to physiological activation? A 1H functional MR spectroscopy study during motor activation in human brain at 7 Tesla** *NeuroImage* **93**:138–145 <https://doi.org/10.1016/j.neuroimage.2014.02.016>
- Siesjö BK, Kogure K., Siesjö K-AH BK, Welsh FA (1985) **Acid-Base Homeostasis in the Brain: Physiology, Chemistry, and Neurochemical Pathology** *Progress in Brain Research, Molecular Mechanisms of Ischemic Brain Damage* :121–154
- Siesjö BK, von Hanwehr R, Nergelius G, Nevander G, Ingvar M. (1985) **Extra- and intracellular pH in the brain during seizures and in the recovery period following the arrest of seizure activity** *J Cereb Blood Flow Metab* **5**:47–57 <https://doi.org/10.1038/jcbfm.1985.7>
- Soeiro-de-Souza MG, Pastorello BF, Leite C da C, Henning A, Moreno RA, Garcia Otaduy MC (2016) **Dorsal anterior cingulate lactate and glutathione levels in euthymic bipolar I disorder: 1H-MRS study** *Int J Neuropsychopharmacol* **19**:1–8 <https://doi.org/10.1093/ijnp/pyw032>
- Stork C, Renshaw PF (2005) **Mitochondrial dysfunction in bipolar disorder: evidence from magnetic resonance spectroscopy research** *Mol Psychiatry* **10**:900–919 <https://doi.org/10.1038/sj.mp.4001711>
- Sullivan CR, Mielnik CA, Funk A, O'Donovan SM, Bentea E, Pletnikov M, Ramsey AJ, Wen Z, Rowland LM, McCullumsmith RE (2019) **Measurement of lactate levels in postmortem brain, iPSCs, and animal models of schizophrenia** *Sci Rep* **9** <https://doi.org/10.1038/s41598-019-41572-9>
- Suzuki A, Stern SA, Bozdagi O, Huntley GW, Walker RH, Magistretti PJ, Alberini CM (2011) **Astrocyte-neuron lactate transport is required for long-term memory formation** *Cell* **144**:810–823 <https://doi.org/10.1016/j.cell.2011.02.018>
- Takao K *et al.* (2013) **Deficiency of Schnurri-2, an MHC enhancer binding protein, induces mild chronic inflammation in the brain and confers molecular, neuronal, and behavioral phenotypes related to schizophrenia** *Neuropsychopharmacology* **38**:1409–1425 <https://doi.org/10.1038/npp.2013.38>
- Takao K, Miyakawa T (2006) **Investigating gene-to-behavior pathways in psychiatric disorders** *Ann N Y Acad Sci* **1086**:144–159 <https://doi.org/10.1196/annals.1377.008>

- Theparambil SM, Hosford PS, Ruminot I, Kopach O, Reynolds JR, Sandoval PY, Rusakov DA, Barros LF, Gourine AV (2020) **Astrocytes regulate brain extracellular pH via a neuronal activity-dependent bicarbonate shuttle** *Nat Commun* **11** <https://doi.org/10.1038/s41467-020-18756-3>
- Tomita H *et al.* (2004) **Effect of agonal and postmortem factors on gene expression profile: quality control in microarray analyses of postmortem human brain** *Biol Psychiatry* **55**:346–352 <https://doi.org/10.1016/j.biopsych.2003.10.013>
- Vawter M *et al.* (2006) **Mitochondrial-related gene expression changes are sensitive to agonal-pH state: implications for brain disorders** *Mol Psychiatry* **11**:615–679 <https://doi.org/10.1038/sj.mp.4001830>
- Weaver KE, Richards TL, Logsdon RG, McGough EL, Minoshima S, Aylward EH, Kleinhans NM, Grabowski TJ, McCurry SM, Teri L (2015) **Posterior cingulate lactate as a metabolic biomarker in amnesic mild cognitive impairment** *BioMed Res Int* **2015** <https://doi.org/10.1155/2015/610605>
- Wenderski W *et al.* (2020) **Loss of the neural-specific BAF subunit ACTL6B relieves repression of early response genes and causes recessive autism** *Proc Natl Acad Sci* **117**:10055–10066 <https://doi.org/10.1073/pnas.1908238117>
- Whitfield-Gabrieli S *et al.* (2009) **Hyperactivity and hyperconnectivity of the default network in schizophrenia and in first-degree relatives of persons with schizophrenia** *Proc Natl Acad Sci* **106**:1279–1284 <https://doi.org/10.1073/pnas.0809141106>
- WHO International Consortium in Psychiatric Epidemiology (2000) **Cross-national comparisons of the prevalences and correlates of mental disorders** *Bull World Health Organ* **78**:413–426
- Won H *et al.* (2012) **Autistic-like social behaviour in Shank2-mutant mice improved by restoring NMDA receptor function** *Nature* **486**:261–265 <https://doi.org/10.1038/nature11208>
- Yao S *et al.* (2023) **Astrocytic lactate dehydrogenase A regulates neuronal excitability and depressive-like behaviors through lactate homeostasis in mice** *Nat Commun* **14** <https://doi.org/10.1038/s41467-023-36209-5>
- Yizhar O *et al.* (2011) **Neocortical excitation/inhibition balance in information processing and social dysfunction** *Nature* **477**:171–178 <https://doi.org/10.1038/nature10360>
- Youssef P, Chami B, Lim J, Middleton T, Sutherland GT, Witting PK (2018) **Evidence supporting oxidative stress in a moderately affected area of the brain in Alzheimer's disease** *Sci Rep* **8**:1–14 <https://doi.org/10.1038/s41598-018-29770-3>
- Zauner A, Bullock R, Di X, Young HF. (1995) **Brain oxygen, CO₂, pH, and temperature monitoring: evaluation in the feline brain** *Neurosurgery* **37**:1168–1177 <https://doi.org/10.1227/00006123-199512000-00017>

Article and author information

Hideo Hagihara

Division of Systems Medical Science, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan

For correspondence: h-hagi@fujita-hu.ac.jp

ORCID iD: [0000-0001-9602-9518](https://orcid.org/0000-0001-9602-9518)

Hiroataka Shoji

Division of Systems Medical Science, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan

ORCID iD: [0000-0003-4843-6949](https://orcid.org/0000-0003-4843-6949)

Satoko Hattori

Division of Systems Medical Science, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan

Giovanni Sala

Division of Systems Medical Science, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan

ORCID iD: [0000-0002-1589-3759](https://orcid.org/0000-0002-1589-3759)

Yoshihiro Takamiya

Division of Systems Medical Science, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan

Mika Tanaka

Division of Systems Medical Science, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan

Masafumi Ihara

Department of Neurology, National Cerebral and Cardiovascular Center, Suita, Osaka 564-8565, Japan

ORCID iD: [0000-0002-7102-4048](https://orcid.org/0000-0002-7102-4048)

Mihiro Shibutani

Laboratory of Genome Science, Biosignal Genome Resource Center, Institute for Molecular and Cellular Regulation, Gunma University, Maebashi, Gunma 371-8512, Japan

Izuho Hatada

Laboratory of Genome Science, Biosignal Genome Resource Center, Institute for Molecular and Cellular Regulation, Gunma University, Maebashi, Gunma 371-8512, Japan

Kei Hori

Department of Biochemistry and Cellular Biology, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan

Mikio Hoshino

Department of Biochemistry and Cellular Biology, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan

Akito Nakao

Department of Synthetic Chemistry and Biological Chemistry, Graduate School of Engineering,
Kyoto University, Kyoto, Kyoto 615-8510, Japan
ORCID iD: [0000-0002-2296-8245](https://orcid.org/0000-0002-2296-8245)

Yasuo Mori

Department of Synthetic Chemistry and Biological Chemistry, Graduate School of Engineering,
Kyoto University, Kyoto, Kyoto 615-8510, Japan

Shigeo Okabe

Department of Cellular Neurobiology, Graduate School of Medicine, The University of Tokyo,
Tokyo 113-0033, Japan
ORCID iD: [0000-0003-1216-8890](https://orcid.org/0000-0003-1216-8890)

Masayuki Matsushita

Department of Molecular and Cellular Physiology, Graduate School of Medicine, University of
the Ryukyus, Nishihara, Okinawa 903-0215, Japan

Anja Urbach

Hans Berger Department of Neurology, Jena University Hospital, Jena D07747, Germany
ORCID iD: [0000-0002-9321-829X](https://orcid.org/0000-0002-9321-829X)

Yuta Katayama

Department of Molecular and Cellular Biology, Medical Institute of Bioregulation, Kyushu
University, Fukuoka, Fukuoka 812-8582, Japan

Akinobu Matsumoto

Department of Molecular and Cellular Biology, Medical Institute of Bioregulation, Kyushu
University, Fukuoka, Fukuoka 812-8582, Japan

Keiichi I. Nakayama

Department of Molecular and Cellular Biology, Medical Institute of Bioregulation, Kyushu
University, Fukuoka, Fukuoka 812-8582, Japan
ORCID iD: [0000-0002-7185-1529](https://orcid.org/0000-0002-7185-1529)

Shota Katori

Laboratory of Mammalian Neural Circuits, National Institute of Genetics, Mishima, Shizuoka
411-8540, Japan

Takuya Sato

Laboratory of Mammalian Neural Circuits, National Institute of Genetics, Mishima, Shizuoka
411-8540, Japan

Takuji Iwasato

Laboratory of Mammalian Neural Circuits, National Institute of Genetics, Mishima, Shizuoka
411-8540, Japan

Haruko Nakamura

Department of Molecular Pharmacology and Neurobiology, Yokohama City University
Graduate School of Medicine, Yokohama, Kanagawa 236-0004, Japan

Yoshio Goshima

Department of Molecular Pharmacology and Neurobiology, Yokohama City University
Graduate School of Medicine, Yokohama, Kanagawa 236-0004, Japan
ORCID iD: [0000-0002-1944-7051](https://orcid.org/0000-0002-1944-7051)

Matthieu Raveau

Laboratory for Neurogenetics, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan

Tetsuya Tatsukawa

Laboratory for Neurogenetics, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan
ORCID iD: [0000-0001-8533-4576](https://orcid.org/0000-0001-8533-4576)

Kazuhiro Yamakawa

Laboratory for Neurogenetics, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan,
Department of Neurodevelopmental Disorder Genetics, Institute of Brain Sciences, Nagoya City
University Graduate School of Medical Sciences, Nagoya, Aichi 467-8601, Japan
ORCID iD: [0000-0002-1478-4390](https://orcid.org/0000-0002-1478-4390)

Noriko Takahashi

Laboratory of Structural Physiology, Center for Disease Biology and Integrative Medicine,
Faculty of Medicine, The University of Tokyo, Tokyo 113-0033, Japan, Department of Physiology,
Kitasato University School of Medicine, Sagamihara, Kanagawa 252-0374, Japan
ORCID iD: [0000-0003-1025-1690](https://orcid.org/0000-0003-1025-1690)

Haruo Kasai

Laboratory of Structural Physiology, Center for Disease Biology and Integrative Medicine,
Faculty of Medicine, The University of Tokyo, Tokyo 113-0033, Japan, International Research
Center for Neurointelligence (WPI-IRCN), UTIAS, The University of Tokyo, Tokyo 113-0033, Japan
ORCID iD: [0000-0003-2327-9027](https://orcid.org/0000-0003-2327-9027)

Johji Inazawa

Research Core Center, Tokyo Medical and Dental University, Tokyo 113-8510, Japan
ORCID iD: [0000-0002-3945-2800](https://orcid.org/0000-0002-3945-2800)

Ikuo Nobuhisa

Department of Stem Cell Regulation, Medical Research Institute, Tokyo Medical and Dental
University, Tokyo 113-8510, Japan

Tetsushi Kagawa

Department of Stem Cell Regulation, Medical Research Institute, Tokyo Medical and Dental
University, Tokyo 113-8510, Japan

Tetsuya Taga

Department of Stem Cell Regulation, Medical Research Institute, Tokyo Medical and Dental
University, Tokyo 113-8510, Japan

Mohamed Darwish

Department of Behavioral Physiology, Graduate School of Innovative Life Science, University of
Toyama, Toyama, Toyama 930-0194, Japan, Department of Biochemistry, Faculty of Pharmacy,
Cairo University, Cairo 11562, Egypt

Hirofumi Nishizono

Medical Research Institute, Kanazawa Medical University, Uchinada, Kahoku, Ishikawa 920-0293, Japan

ORCID iD: [0000-0002-9394-7089](https://orcid.org/0000-0002-9394-7089)

Keizo Takao

Department of Behavioral Physiology, Graduate School of Innovative Life Science, University of Toyama, Toyama, Toyama 930-0194, Japan, Life Science Research Center, University of Toyama, Toyama, Toyama 930-0194, Japan

ORCID iD: [0000-0002-4734-3583](https://orcid.org/0000-0002-4734-3583)

Kiran Sapkota

Department of Neuroscience, Southern Research, Birmingham, AL 35205, USA

ORCID iD: [0000-0003-3737-2900](https://orcid.org/0000-0003-3737-2900)

Kazutoshi Nakazawa

Department of Neuroscience, Southern Research, Birmingham, AL 35205, USA

ORCID iD: [0000-0001-5699-9093](https://orcid.org/0000-0001-5699-9093)

Tsuyoshi Takagi

Institute for Developmental Research, Aichi Developmental Disability Center, Kasugai, Aichi 480-0392, Japan

Haruki Fujisawa

Department of Endocrinology, Diabetes and Metabolism, School of Medicine, Fujita Health University, Toyoake, Aichi 470-1192, Japan

Yoshihisa Sugimura

Department of Endocrinology, Diabetes and Metabolism, School of Medicine, Fujita Health University, Toyoake, Aichi 470-1192, Japan

Kyosuke Yamanishi

Department of Neuropsychiatry, Hyogo Medical University, School of Medicine, Nishinomiya, Hyogo 663-8501, Japan

Lakshmi Rajagopal

Department of Psychiatry and Behavioral Sciences, Northwestern University Feinberg School of Medicine, Chicago, IL 60611, USA

ORCID iD: [0000-0002-2382-258X](https://orcid.org/0000-0002-2382-258X)

Nanette Deneen Hannah

Department of Psychiatry and Behavioral Sciences, Northwestern University Feinberg School of Medicine, Chicago, IL 60611, USA

Herbert Y. Meltzer

Department of Psychiatry and Behavioral Sciences, Northwestern University Feinberg School of Medicine, Chicago, IL 60611, USA

ORCID iD: [0000-0003-4901-8845](https://orcid.org/0000-0003-4901-8845)

Tohru Yamamoto

Department of Molecular Neurobiology, Faculty of Medicine, Kagawa University, Kita-gun, Kagawa 761-0793, Japan

ORCID iD: [0000-0002-1652-0233](https://orcid.org/0000-0002-1652-0233)

Shuji Wakatsuki

Department of Peripheral Nervous System Research, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan

ORCID iD: [0000-0003-2403-5876](https://orcid.org/0000-0003-2403-5876)

Toshiyuki Araki

Department of Peripheral Nervous System Research, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan

ORCID iD: [0000-0003-3625-2042](https://orcid.org/0000-0003-3625-2042)

Katsuhiko Tabuchi

Department of Molecular & Cellular Physiology, Shinshu University School of Medicine, Matsumoto, Nagano 390-8621, Japan

ORCID iD: [0000-0002-1673-4637](https://orcid.org/0000-0002-1673-4637)

Tadahiro Numakawa

Department of Mental Disorder Research, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan

Hiroshi Kunugi

Department of Mental Disorder Research, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan, Department of Psychiatry, Teikyo University School of Medicine, Tokyo 173-8605, Japan

ORCID iD: [0000-0002-7209-3790](https://orcid.org/0000-0002-7209-3790)

Freesia L. Huang

Program of Developmental Neurobiology, National Institute of Child Health and Human Development, National Institute of Health, Bethesda, MD 20892, USA

Atsuko Hayata-Takano

Laboratory of Molecular Neuropharmacology, Graduate School of Pharmaceutical Sciences, Osaka University, Suita, Osaka 565-0871, Japan, Department of Pharmacology, Graduate School of Dentistry, Osaka University, Suita, Osaka 565-0871, Japan, United Graduate School of Child Development, Osaka University, Kanazawa University, Hamamatsu University School of Medicine, Chiba University and University of Fukui, Suita, Osaka 565-0871, Japan

ORCID iD: [0000-0002-4992-2845](https://orcid.org/0000-0002-4992-2845)

Hitoshi Hashimoto

Laboratory of Molecular Neuropharmacology, Graduate School of Pharmaceutical Sciences, Osaka University, Suita, Osaka 565-0871, Japan, United Graduate School of Child Development, Osaka University, Kanazawa University, Hamamatsu University School of Medicine, Chiba University and University of Fukui, Suita, Osaka 565-0871, Japan, Division of Bioscience, Institute for Data Biology Science, Osaka University, Suita, Osaka 565-0871, Japan, Transdimensional Life Imaging Division, Institute for Open and Transdisciplinary Research Initiatives, Osaka University, Suita, Osaka 565-0871, Japan, Department of Molecular Pharmaceutical Science, Graduate School of Medicine, Osaka University, Suita, Osaka 565-0871, Japan

ORCID iD: [0000-0001-6548-4016](https://orcid.org/0000-0001-6548-4016)

Kota Tamada

RIKEN Brain Science Institute, Wako, Saitama 351-0198, Japan, Department of Physiology and Cell Biology, Kobe University School of Medicine, Kobe, Hyogo 650-0017, Japan

Toru Takumi

RIKEN Brain Science Institute, Wako, Saitama 351-0198, Japan, Department of Physiology and Cell Biology, Kobe University School of Medicine, Kobe, Hyogo 650-0017, Japan

ORCID iD: [0000-0001-7153-266X](https://orcid.org/0000-0001-7153-266X)

Takaoki Kasahara

Laboratory for Molecular Dynamics of Mental Disorders, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan, Institute of Biology and Environmental Sciences, Carl von Ossietzky University of Oldenburg, Oldenburg 26129, Germany

ORCID iD: [0000-0002-0953-5146](https://orcid.org/0000-0002-0953-5146)

Tadafumi Kato

Laboratory for Molecular Dynamics of Mental Disorders, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan, Department of Psychiatry and Behavioral Science, Juntendo University Graduate School of Medicine, Tokyo 113-8421, Japan

ORCID iD: [0000-0001-7856-3952](https://orcid.org/0000-0001-7856-3952)

Isabella A. Graef

Department of Pathology, Stanford University School of Medicine, Stanford, CA 94305, USA

Gerald R. Crabtree

Department of Pathology, Stanford University School of Medicine, Stanford, CA 94305, USA

Nozomi Asaoka

Department of Pharmacology, Kyoto Prefectural University of Medicine, Kyoto, Kyoto 602-8566, Japan

ORCID iD: [0000-0003-4854-2100](https://orcid.org/0000-0003-4854-2100)

Hikari Hatakama

Department of Molecular Pharmacology, Graduate School of Pharmaceutical Sciences, Kyoto University, Kyoto, Kyoto 606-8501, Japan

Shuji Kaneko

Department of Molecular Pharmacology, Graduate School of Pharmaceutical Sciences, Kyoto University, Kyoto, Kyoto 606-8501, Japan
ORCID iD: [0000-0001-5152-5809](https://orcid.org/0000-0001-5152-5809)

Takao Kohno

Department of Biomedical Science, Graduate School of Pharmaceutical Sciences, Nagoya City University, Nagoya, Aichi 467-8603, Japan
ORCID iD: [0000-0003-2557-4281](https://orcid.org/0000-0003-2557-4281)

Mitsuharu Hattori

Department of Biomedical Science, Graduate School of Pharmaceutical Sciences, Nagoya City University, Nagoya, Aichi 467-8603, Japan
ORCID iD: [0000-0002-0503-4969](https://orcid.org/0000-0002-0503-4969)

Yoshio Hoshiba

Laboratory of Medical Neuroscience, Institute for Molecular and Cellular Regulation, Gunma University, Maebashi, Gunma 371-8512, Japan

Ryuhei Miyake

Laboratory for Multi-scale Biological Psychiatry, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan

Kisho Obi-Nagata

Laboratory for Multi-scale Biological Psychiatry, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan

Akiko Hayashi-Takagi

Laboratory of Medical Neuroscience, Institute for Molecular and Cellular Regulation, Gunma University, Maebashi, Gunma 371-8512, Japan, Laboratory for Multi-scale Biological Psychiatry, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan

Léa J. Becker

Institut des Neurosciences Cellulaires et Intégratives, Centre National de la Recherche Scientifique, Université de Strasbourg, Strasbourg 67000, France

Ipek Yalcin

Institut des Neurosciences Cellulaires et Intégratives, Centre National de la Recherche Scientifique, Université de Strasbourg, Strasbourg 67000, France
ORCID iD: [0000-0003-2596-8598](https://orcid.org/0000-0003-2596-8598)

Yoko Hagino

Addictive Substance Project, Tokyo Metropolitan Institute of Medical Science, Tokyo 156-8506, Japan

Hiroko Kotajima-Murakami

Addictive Substance Project, Tokyo Metropolitan Institute of Medical Science, Tokyo 156-8506, Japan

Yuki Moriya

Addictive Substance Project, Tokyo Metropolitan Institute of Medical Science, Tokyo 156-8506, Japan

Kazutaka Ikeda

Addictive Substance Project, Tokyo Metropolitan Institute of Medical Science, Tokyo 156-8506, Japan

ORCID iD: [0000-0001-8342-0278](https://orcid.org/0000-0001-8342-0278)

Hyopil Kim

Department of Biological Sciences, College of Natural Sciences, Seoul National University, Seoul 08826, South Korea

ORCID iD: [0000-0001-9159-8682](https://orcid.org/0000-0001-9159-8682)

Bong-Kiun Kaang

Department of Biological Sciences, College of Natural Sciences, Seoul National University, Seoul 08826, South Korea

ORCID iD: [0000-0001-7593-9707](https://orcid.org/0000-0001-7593-9707)

Hikari Otabi

College of Agriculture, Ibaraki University, Ami, Ibaraki 300-0393, Japan, United Graduate School of Agricultural Science, Tokyo University of Agriculture and Technology, Fuchu, Tokyo 183-8538, Japan

ORCID iD: [0000-0001-5127-8190](https://orcid.org/0000-0001-5127-8190)

Yuta Yoshida

College of Agriculture, Ibaraki University, Ami, Ibaraki 300-0393, Japan

Atsushi Toyoda

College of Agriculture, Ibaraki University, Ami, Ibaraki 300-0393, Japan, United Graduate School of Agricultural Science, Tokyo University of Agriculture and Technology, Fuchu, Tokyo 183-8538, Japan, Ibaraki University Cooperation between Agriculture and Medical Science (IUCAM), Ami, Ibaraki 300-0393, Japan

ORCID iD: [0000-0002-0245-3244](https://orcid.org/0000-0002-0245-3244)

Noboru H. Komiyama

Genes to Cognition Program, Centre for Clinical Brain Sciences, University of Edinburgh, Edinburgh EH16 4SB, UK, Simons Initiative for the Developing Brain, Centre for Discovery Brain Sciences, University of Edinburgh, Edinburgh EH8 9XF, UK

ORCID iD: [0000-0001-9960-3597](https://orcid.org/0000-0001-9960-3597)

Seth G. N. Grant

Genes to Cognition Program, Centre for Clinical Brain Sciences, University of Edinburgh, Edinburgh EH16 4SB, UK, Simons Initiative for the Developing Brain, Centre for Discovery Brain Sciences, University of Edinburgh, Edinburgh EH8 9XF, UK

ORCID iD: [0000-0001-8732-8735](https://orcid.org/0000-0001-8732-8735)

Michiru Ida-Eto

Department of Developmental and Regenerative Medicine, Mie University, Graduate School of Medicine, Tsu, Mie 514-8507, Japan

ORCID iD: [0000-0002-2469-4006](https://orcid.org/0000-0002-2469-4006)

Masaaki Narita

Department of Developmental and Regenerative Medicine, Mie University, Graduate School of Medicine, Tsu, Mie 514-8507, Japan

Ken-ichi Matsumoto

Department of Biosignaling and Radioisotope Experiment, Interdisciplinary Center for Science Research, Organization for Research and Academic Information, Shimane University, Izumo, Shimane 693-8501, Japan

Emiko Okuda-Ashitaka

Department of Biomedical Engineering, Osaka Institute of Technology, Osaka 535-8585, Japan

Iori Ohmori

Department of Physiology, Okayama University Graduate School of Medicine, Dentistry and Pharmaceutical Sciences, Okayama, Okayama 700-8530, Japan
ORCID iD: [0000-0001-6823-2762](https://orcid.org/0000-0001-6823-2762)

Tadayuki Shimada

Child brain project, Tokyo Metropolitan Institute of Medical Science, Tokyo 156-8506, Japan

Kanato Yamagata

Child brain project, Tokyo Metropolitan Institute of Medical Science, Tokyo 156-8506, Japan

Hiroshi Ageta

Division for Therapies Against Intractable Diseases, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan

Kunihiro Tsuchida

Division for Therapies Against Intractable Diseases, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan
ORCID iD: [0000-0002-3983-5756](https://orcid.org/0000-0002-3983-5756)

Kaoru Inokuchi

Research Center for Idling Brain Science, University of Toyama, Toyama, Toyama 930-0194, Japan, Department of Biochemistry, Graduate School of Medicine and Pharmaceutical Sciences, University of Toyama, Toyama, Toyama 930-0194, Japan, Core Research for Evolutionary Science and Technology (CREST), Japan Science and Technology Agency (JST), University of Toyama, Toyama, Toyama 930-0194, Japan
ORCID iD: [0000-0002-5393-3133](https://orcid.org/0000-0002-5393-3133)

Takayuki Sassa

Faculty of Pharmaceutical Sciences, Hokkaido University, Sapporo, Hokkaido 060-0812, Japan
ORCID iD: [0000-0003-3145-9829](https://orcid.org/0000-0003-3145-9829)

Akio Kihara

Faculty of Pharmaceutical Sciences, Hokkaido University, Sapporo, Hokkaido 060-0812, Japan
ORCID iD: [0000-0001-5889-0788](https://orcid.org/0000-0001-5889-0788)

Motoaki Fukasawa

Department of Anatomy II, Fujita Health University School of Medicine, Toyoake, Aichi 470-1192, Japan

ORCID iD: [0000-0002-2000-9446](https://orcid.org/0000-0002-2000-9446)

Nobuteru Usuda

Department of Anatomy II, Fujita Health University School of Medicine, Toyoake, Aichi 470-1192, Japan

Tayo Katano

Department of Medical Chemistry, Kansai Medical University, Hirakata, Osaka 573-1010, Japan

ORCID iD: [0000-0002-7969-744X](https://orcid.org/0000-0002-7969-744X)

Teruyuki Tanaka

Department of Developmental Medical Sciences, Graduate School of Medicine, The University of Tokyo, Tokyo 113-0033, Japan

ORCID iD: [0000-0003-2000-5354](https://orcid.org/0000-0003-2000-5354)

Yoshihiro Yoshihara

Laboratory for Systems Molecular Ethology, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan

ORCID iD: [0000-0002-0319-7166](https://orcid.org/0000-0002-0319-7166)

Michihiro Igarashi

Department of Neurochemistry and Molecular Cell Biology, Niigata University Graduate School of Medical and Dental Sciences, Niigata, Niigata 951-8510, Japan, Transdisciplinary Research Program, Niigata University, Niigata, Niigata 951-8510, Japan

Takashi Hayashi

Biomedical Research Institute, National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Ibaraki 305-8566, Japan

ORCID iD: [0000-0003-3591-2109](https://orcid.org/0000-0003-3591-2109)

Kaori Ishikawa

Faculty of Life and Environmental Sciences, University of Tsukuba, Tsukuba, Ibaraki 305-8572, Japan, Graduate School of Life and Environmental Sciences, University of Tsukuba, Tsukuba, Ibaraki 305-8572, Japan

ORCID iD: [0000-0001-9895-5996](https://orcid.org/0000-0001-9895-5996)

Satoshi Yamamoto

Integrated Technology Research Laboratories, Pharmaceutical Research Division, Takeda Pharmaceutical Company, Ltd, Fujisawa, Kanagawa 251-8555, Japan

Naoya Nishimura

Integrated Technology Research Laboratories, Pharmaceutical Research Division, Takeda Pharmaceutical Company, Ltd, Fujisawa, Kanagawa 251-8555, Japan

Kazuto Nakada

Faculty of Life and Environmental Sciences, University of Tsukuba, Tsukuba, Ibaraki 305-8572, Japan, Graduate School of Life and Environmental Sciences, University of Tsukuba, Tsukuba, Ibaraki 305-8572, Japan

Shinji Hirotsume

Department of Genetic Disease Research, Osaka City University Graduate School of Medicine, Osaka, Osaka 545-8585, Japan

Kiyoshi Egawa

Department of Pediatrics, Hokkaido University Graduate School of Medicine, Sapporo, Hokkaido 060-8638, Japan

Kazuma Higashisaka

Laboratory of Toxicology and Safety Science, Graduate School of Pharmaceutical Sciences, Osaka University, Suita, Osaka 565-0871, Japan

Yasuo Tsutsumi

Laboratory of Toxicology and Safety Science, Graduate School of Pharmaceutical Sciences, Osaka University, Suita, Osaka 565-0871, Japan

Shoko Nishihara

Glycan & Life Systems Integration Center (GaLSIC), Soka University, Tokyo 192-8577, Japan

Noriyuki Sugo

Graduate School of Frontier Biosciences, Osaka University, Suita, Osaka 565-0871, Japan

Takeshi Yagi

Graduate School of Frontier Biosciences, Osaka University, Suita, Osaka 565-0871, Japan

Naoto Ueno

National Institute for Basic Biology, Laboratory of Morphogenesis, Okazaki, Aichi 444-8585, Japan
ORCID iD: [0000-0002-8375-2317](https://orcid.org/0000-0002-8375-2317)

Tomomi Yamamoto

Division of Biophysics and Neurobiology, National Institute for Physiological Sciences, Okazaki, Aichi 444-8585, Japan

Yoshihiro Kubo

Division of Biophysics and Neurobiology, National Institute for Physiological Sciences, Okazaki, Aichi 444-8585, Japan
ORCID iD: [0000-0001-6707-0837](https://orcid.org/0000-0001-6707-0837)

Rie Ohashi

Laboratory of Neuronal Cell Biology, National Institute for Basic Biology, Okazaki, Aichi 444-8585, Japan, Department of Basic Biology, SOKENDAI (Graduate University for Advanced Studies), Okazaki, Aichi 444-8585, Japan, Exploratory Research Center on Life and Living Systems (ExCELLS), Okazaki, Aichi 444-8585, Japan

Nobuyuki Shiina

Laboratory of Neuronal Cell Biology, National Institute for Basic Biology, Okazaki, Aichi 444-8585, Japan, Department of Basic Biology, SOKENDAI (Graduate University for Advanced Studies), Okazaki, Aichi 444-8585, Japan, Exploratory Research Center on Life and Living Systems (ExCELLS), Okazaki, Aichi 444-8585, Japan
ORCID iD: [0000-0002-1854-4239](https://orcid.org/0000-0002-1854-4239)

Kimiko Shimizu

Department of Biological Sciences, School of Science, The University of Tokyo, Tokyo 113-0033, Japan

ORCID iD: [0000-0003-4943-2552](https://orcid.org/0000-0003-4943-2552)

Sayaka Higo-Yamamoto

Healthy Food Science Research Group, Cellular and Molecular Biotechnology Research Institute, National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Ibaraki 305-8566, Japan

Katsutaka Oishi

Healthy Food Science Research Group, Cellular and Molecular Biotechnology Research Institute, National Institute of Advanced Industrial Science and Technology (AIST), Tsukuba, Ibaraki 305-8566, Japan, Department of Applied Biological Science, Graduate School of Science and Technology, Tokyo University of Science, Noda, Chiba 278-8510, Japan, Department of Computational Biology and Medical Sciences, Graduate School of Frontier Sciences, The University of Tokyo, Kashiwa, Chiba 277-8562, Japan, School of Integrative and Global Majors (SIGMA), University of Tsukuba, Tsukuba, Ibaraki 305-8577, Japan

ORCID iD: [0000-0002-4870-0837](https://orcid.org/0000-0002-4870-0837)

Hisashi Mori

Department of Molecular Neuroscience, Graduate School of Medicine and Pharmaceutical Sciences, University of Toyama, Toyama, Toyama 930-0194, Japan

ORCID iD: [0000-0001-9743-2456](https://orcid.org/0000-0001-9743-2456)

Tamio Furuse

Technology and Development Team for Mouse Phenotype Analysis, Japan Mouse Clinic, RIKEN BioResource Research Center (BRC), Tsukuba, Ibaraki 305-0074, Japan

ORCID iD: [0000-0002-9276-6040](https://orcid.org/0000-0002-9276-6040)

Masaru Tamura

Technology and Development Team for Mouse Phenotype Analysis, Japan Mouse Clinic, RIKEN BioResource Research Center (BRC), Tsukuba, Ibaraki 305-0074, Japan

Hisashi Shirakawa

Department of Molecular Pharmacology, Graduate School of Pharmaceutical Sciences, Kyoto University, Kyoto, Kyoto 606-8501, Japan

ORCID iD: [0000-0002-4129-0978](https://orcid.org/0000-0002-4129-0978)

Daiki X. Sato

Division of Systems Medical Science, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan, Graduate School of Life Sciences, Tohoku University, Sendai, Miyagi 980-8578, Japan

ORCID iD: [0000-0002-9527-8253](https://orcid.org/0000-0002-9527-8253)

Yukiko U. Inoue

Department of Biochemistry and Cellular Biology, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan

ORCID iD: [0000-0002-4105-9127](https://orcid.org/0000-0002-4105-9127)

Takayoshi Inoue

Department of Biochemistry and Cellular Biology, National Institute of Neuroscience, National Center of Neurology and Psychiatry, Kodaira, Tokyo 187-8502, Japan
ORCID iD: [0000-0001-7155-2150](https://orcid.org/0000-0001-7155-2150)

Yuriko Komine

Young Researcher Support Group, Research Enhancement Strategy Office, National Institute for Basic Biology, National Institute of Natural Sciences, Okazaki, Aichi 444-8585, Japan
ORCID iD: [0000-0002-0097-8813](https://orcid.org/0000-0002-0097-8813)

Tetsuo Yamamori

Division of Brain Biology, National Institute for Basic Biology, Okazaki, Aichi 444-8585, Japan, Laboratory for Molecular Analysis of Higher Brain Function, RIKEN Center for Brain Science, Wako, Saitama 351-0198, Japan

Kenji Sakimura

Department of Cellular Neurobiology, Brain Research Institute, Niigata University, Niigata, Niigata 951-8585, Japan, Department of Animal Model Development, Brain Research Institute, Niigata University, Niigata, Niigata 951-8585, Japan
ORCID iD: [0000-0002-8091-8879](https://orcid.org/0000-0002-8091-8879)

Tsuyoshi Miyakawa

Division of Systems Medical Science, Center for Medical Science, Fujita Health University, Toyoake, Aichi 470-1192, Japan
For correspondence: miyakawa@fujita-hu.ac.jp
ORCID iD: [0000-0003-0137-8200](https://orcid.org/0000-0003-0137-8200)

Copyright

© 2023, Hagihara et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Michelle Antoine

National Institute on Alcohol Abuse and Alcoholism (NIAAA), Bethesda, United States of America

Senior Editor

Sacha Nelson

Brandeis University, Waltham, United States of America

Reviewer #1 (Public Review):**Summary**

In this manuscript, Hagihara et al. characterized the relationship between the changes in lactate and pH and the behavioral phenotypes in different animal models of neuropsychiatric disorders at a large-scale level. The authors have previously reported that increased lactate levels and decreased pH are commonly observed in the brains of five genetic mouse models

of schizophrenia (SZ), bipolar disorder (BD), and autism spectrum disorder (ASD). In this study, they expanded the detection range to 109 strains or conditions of animal models, covering neuropsychiatric disorders and neurodegenerative disorders. Through statistical analysis of the first 65 strains/conditions of animal models which were set as exploratory cohort, the authors found that most strains showed decreased pH and increased lactate levels in the brains. There was a significant negative correlation between pH and lactate levels both at the strain/condition level and the individual animal level. Besides, only working memory was negatively correlated with brain lactate levels. These results were successfully duplicated by studying the confirmative cohort, including 44 strains/conditions of animal models. In all strains/conditions, the lactate levels were not correlated with age, sex, or storage duration of brain samples.

Strengths

1. The manuscript is well-written and structured. In particular, the discussion is really nice, covering many potential mechanisms for the altered lactate levels in these disease models.
2. Tremendous efforts were made to recruit a huge number of various animal models, giving the conclusions sufficient power.

Weaknesses

1. The biggest concern of this study is the limited novelty. The point of "altered pH and/or lactate levels in the brains from human and rodent animals of neuropsychiatric disorders" has been reported by the same lab and other groups in many previous papers.
2. This study is mostly descriptive, lacking functional investigations. Although a larger cohort of animal models were studied which makes the conclusion more solid, limited conceptual advance is contributed to the relevant field, as we are still not clear about what the altered levels of pH and lactate mean for the pathogenesis of neuropsychiatric disorders.
3. The experiment procedure is also a concern. The brains from animal models were acutely collected without cardiac perfusion in this study, which suggests that resident blood may contaminate the brain samples. The lactate is enriched in the blood, making it a potential confounded factor to affect the lactate levels as well as pH in the brain samples.
4. The lactate and pH levels may also be affected by other confounded factors, such as circadian period, and locomotor activity before the mice were sacrificed. This should also be discussed in the paper.
5. Another concern is the animal models. Although previous studies have demonstrated that dysfunctions of these genes could cause related phenotypes for certain disorders, many of them are not acknowledged by the field as reliable disease models. Besides, gene deficiency could also cause many known or unknown unrelated phenotypes, which may contribute to the altered levels of lactate and pH, too. In this circumstance, the conclusion "pH and lactate levels are transdiagnostic endophenotype of neuropsychiatric disorders" is somewhat overstated.
6. The negative relationship between pH and lactate is rather convincing. However, how much the contribution of lactate to pH is not tested. In addition, regarding pH and lactate, which factor contributes most to the pathogenesis of neuropsychiatric disorders is also unclear. These questions may need to be addressed in the future study.
7. The authorship is open to question. Most authors listed in this paper may only provide mice strains or brain samples. Maybe it is better just to acknowledge them in the acknowledgments section.
8. The last concern is about the significance of this study. Although the majority of strains showed increased lactate, some still showed decreased lactate levels in the brains. These results suggested that lactate or pH is an endophenotype for neuropsychiatric disorders, but it is hard to serve as a good diagnostic index as the change is not unidirectional in different disorders. In other words, the relationship between lactate level and neuropsychiatric disorders is not exclusive.

- <https://doi.org/10.7554/eLife.89376.1.sa1>

Reviewer #2 (Public Review):

Hagihara et al. conducted a study investigating the correlation between decreased brain pH, increased brain lactate, and poor working memory. They found altered brain pH and lactate levels in animal models of neuropsychiatric and neurodegenerative disorders. Their study suggests that poor working memory performance may predict higher brain lactate levels.

However, the study has some significant limitations. One major concern is that the authors examined whole-brain pH and lactate levels, which might not fully represent the complexity of disease states. Different brain regions and cell types may have distinct protein and metabolite profiles, leading to diverse disease outcomes. For instance, certain brain regions like the hippocampus and nucleus accumbens exhibit opposite protein/signaling pathways in neuropsychiatric disease models.

Moreover, the memory tests used in the study are specific to certain brain regions, but the authors did not measure lactate levels in those regions. Without making lactate measurements in brain-regions and cell types involved in these diseases, any conclusions regarding the role of lactate in CNS diseases is premature.

Additionally, evidence suggests that exogenous treatment with lactate has positive effects, such as antidepressant effects in multiple disease models (Carrard et al., 2018, Carrard et al., 2021, Karnib et al., 2019, Shaif et al., 2018). It also promotes learning, memory formation, neurogenesis, and synaptic plasticity (Suzuki et al., 2011, Yang et al., 2014, Weitian et al., 2015, Dong et al., 2017, El Hayek et al. 2019, Wang et al., 2019, Lu et al., 2019, Lev-Vachnisch et al., 2019, Descalzi G et al., 2019, Herrera-López et al., 2020, Ikeda et al., 2021, Zhou et al., 2021, Roumes et al., 2021, Frame et al., 2023, Akter et al., 2023).

In conclusion, the relevance of total brain pH and lactate levels as indicators of the observed correlations is controversial, and evidence points towards lactate having more positive rather than negative effects. It is important that the authors perform studies looking at brain-region-specific concentrations of lactate and that they modulate lactate levels (decrease) in animal models of disease to validate their conclusions. It is also important to consider the above-mentioned studies before concluding that "altered brain pH and lactate levels are rather involved in the underlying pathophysiology of some patients with neuropsychiatric disorders" and that "lactate can serve as a potential therapeutic target for neuropsychiatric disorders".

- <https://doi.org/10.7554/eLife.89376.1.sa0>